

THE GEOMETRY OF TRICKLE GROUPS

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ABSTRACT. We introduce the concept of prismatic presentations, a family of presentations designed to have disc diagrams with nice combinatorics. These combinatorics are expressed in terms of certain subdiagrams, corridors and spiders, whose theory we develop to show that we can exclude certain pathological subdiagrams. Our main result is a diagrammatic characterisation of geodesic words. As an application of our main theorem, we prove that groups having prismatic presentations with additional mild assumptions enjoy the falsification by fellow-traveller property, are shortlex automatic or admit Tits rewriting algorithms. We show that examples of groups admitting prismatic presentations include trickle groups (right-angled Artin groups, right-angled Coxeter groups, cactus groups, and more), twisted right-angled Artin groups and Graph products. In particular, we address some questions of Bellingeri, Godelle and Paris about trickle groups.

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1. INTRODUCTION

NEW INTRO

The main purpose of this paper is to provide geometric tools to understand and work with *trickle groups* [4]. For this, we introduce what we call *prismatic presentations*, that provide a more general framework that encompasses (not necessarily trickle) groups with trickle-like geometry. The definition of this type of presentation ensures that its associated disc diagrams exhibit a good geometric behaviour, which we exploit to obtain information about these groups using geometric methods.

Informally, we can say that a presentation $\langle \Sigma | R \rangle$ is prismatic if the set of relators admits a partition $R = S \sqcup H$ such that, in the Cayley complex X associated to $\langle \Sigma | R \rangle$, the following properties hold:

- I1** (Squares and heads) The 2-cells in X associated to S -relators are all squares, whereas the H -relator 2-cells can be polygons of 2 or more sides.
- I2** (Cube completion) If X contains three S -relator squares glued as in the corner of a cube, then the whole cube must be present in X (see Fig. 1(A)).
- I3** (Prism completion) If an S -relator 2-cell and a H -relator 2-cell share exactly one edge, then X contains a prism where one of the bases is the H -relator, one of the lateral faces is the S -relator and they are glued along their overlapping side (see Fig. 1(B)).

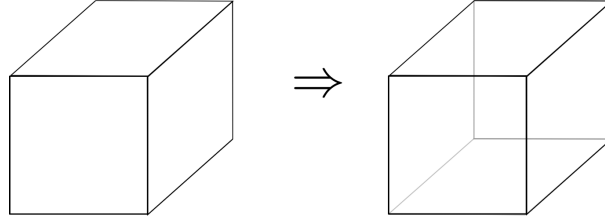
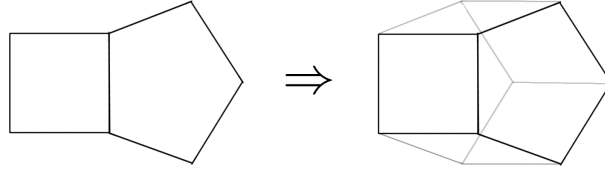
(A) Condition **I2**: cube completion(B) Condition **I3**: prism completion

FIG. 1. Properties of a prismatic presentation

Due to the nature of the definition of a prismatic presentation, it is not clear that building concrete examples of prismatic presentations is easy. However, in Section 6.1 we prove the following:

Theorem A (Theorem 6.14). *The standard presentation of a trickle group is prismatic.*

As a consequence we obtain many examples of groups with a prismatic presentation: the family of trickle groups, introduced in [4], includes cactus groups, right-angled Artin groups, right-angled Coxeter groups and, in general, graph products

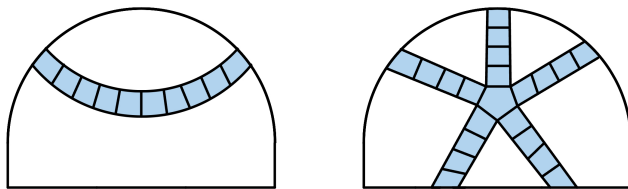


FIG. 2. Corridor (left) and spider (right).

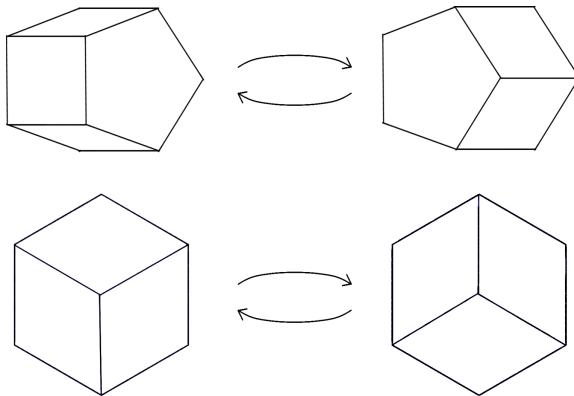


FIG. 3. Prismatic moves (above) and hexagonal moves (below)

of cyclic groups. In addition to trickle groups, in Section 6.2, we see that twisted right-angled Artin groups (TRAAGs) and a generalisation of these admit prismatic presentations. It follows from [2] and [13] that there are twisted right-angled Artin groups which are not trickle groups (Remark 6.15), so the family of prismatic presentations is larger than the family of trickle groups.

The formal definition of prismatic presentations includes another condition regarding the overlap between the different relators (see Section 2), but the conditions stated here tell us already a lot about the disc diagrams over prismatic presentations. Indeed, from Item I1, we may see these diagrams as the union of two types of subdiagrams: *corridors* and *spiders* (see Fig. 2). These will be formally defined in Section 3, but note that corridors have appeared before in the literature (see [7, 23]).

As to conditions I2 and I3, they illustrate the key point of prismatic presentations: any disc diagram over a prismatic presentation admits *hexagonal* and *prismatic moves* (see Fig. 3). This means that, whenever we see a hexagon tessellated by 3 squares in a disc diagram, we can substitute this subdiagram by the alternative tessellation (given by the other side of the cube, which we know exists by the cube completion condition). Similarly, if we see a subdiagram which can be seen as part of a full prism, we can retessellate it by its complement in the prism (using the prism completion condition). Hexagonal moves appear in the literature in the context of non positively curved cube complexes (see [31], for example) and they will also play an important role in our proofs. As for prismatic moves, we introduce them to deal with torsion relations in trickle groups and, in the more general context of prismatic presentations, their main purpose is to “move around”

the H -relator 2-cells in a disc diagram, so they will also be heavily used in our proofs.

For applications, we will impose different extra conditions on the heads of prismatic presentations: saturation (Definition 4.2), cyclic spiderheads (Definition 2.7), triangular spiderheads (Definition 2.7), etc. The standard presentation of trickle groups, our initial object of study, falls in the category of prismatic presentations with cyclic spiderheads.

It is important to note that the results in Section 3 do not require any of these extra assumptions. This will allow us to work with non-pathological diagrams over prismatic presentations, that is, diagrams that do not contain subdiagrams that are technically problematic (for example, a double crossing of two corridors).

Adding the condition of saturation to prismatic presentations, in Section 4, we obtain a geometric characterisation of geodesics over such a presentation:

Theorem B (Theorem 4.23). *Let $\langle \Sigma \mid S \sqcup H \rangle$ be a saturated prismatic presentation, and let w, u be words over its generators representing the same group element. Then, w is geodesic if and only if no disc diagram D for wu^{-1} that is free of pathologies and siamese spiders contains:*

- (1) *A corridor both of whose ends lie on $w \cap \partial D$, or*
- (2) *A spider with the majority of its legs ending on $w \cap \partial D$.*

From this characterisation of geodesics, in Section 5 we obtain several interesting results, adding the convenient hypotheses in each case. For example, we recover a Tits-style rewriting system, described in Definition 5.15, similar to the one proposed for trickle groups in [4, §2.3]. Another application gives an important tool to understand the Cayley graph of prismatic presentations. We say that a graph Γ satisfies the *falsification by fellow-traveller property* (FFTP) if there exists a global constant K such that for every non-geodesic path in Γ , there exists a shorter path with the same endpoints that K -fellow travels the non-geodesic path (that is, it remains uniformly at distance K from the non-geodesic path), and we say that a group G with finite generating set Σ has the FFTP if the Cayley graph of G with respect to Σ satisfies this property. In this case, adding the extra condition of having k -locally excluding spiderheads (see Definition 5.3), we obtain this result:

Theorem C (Theorem 5.4). *If a group G with finite generating set Σ admits a saturated prismatic presentation over Σ with k -locally excluding spiderheads, then the pair (G, Σ) satisfies the falsification by fellow traveller property (FFTP).*

The falsification by fellow traveller property has been shown to hold for many families of groups (see [21, 28, 8, 19, 20, 1]) and it has interesting implications for the groups that have it. Indeed, groups with FFTP are of type F_3 , their Dehn function is at most quadratic (see [9]), they have a regular set of geodesics (see [25]) and, as a consequence, they have a rational growth function.

One further application uses our results in [3], where we show that if a group $\langle \Sigma \rangle$ has the FFTP, then its shortlex representatives form a regular language for any total ordering of Σ . This fact allows us to obtain another useful result for prismatic presentations:

Theorem D (Theorem 5.24). *Every group admitting a finite prismatic presentation with cyclic spiderheads is shortlex automatic.*

This theorem partially answers a question posed in [4] regarding the automaticity of trickle groups. *Automaticity* is a desirable property for groups, as it allows one to study them from an algorithmic point of view. There is some previous work in the literature showing that some subfamilies of prismatic presentations are automatic (see [30, 18] for RAAGs and [11] for TRAAGs; in [13], the author shows that trickle groups are cocompactly cubulable and automaticity can be deduced from this). In our case, by proving that groups with a prismatic presentation are *shortlex* automatic, we provide a very specific automatic structure for this larger class of groups.

Lastly, let us remark that prismatic presentations yield Cayley complexes that can be found to have similarities with other complexes defined in the literature, as is the case of bucolic complexes [5] and quasi-median graphs [24, 12]. Despite the similarities, we note that the Cayley complex of a prismatic presentation does not induce a bucolic complex (because we do not require the *wheel condition* to be satisfied, see [5]) or a quasi-median graph (because we do not require the absence of induced K_4^- graphs, see [13, Theorem 4.4]).

The paper is structured as follows: in Section 2 we give the formal definition of prismatic presentations and spiderhead presentations, as well as an informal introduction to the main examples of prismatic presentations; Section 3 contains the basic notions needed to work with disc diagrams over prismatic presentations (corridors, spiders, pathologies) leading to a result that allows us to work with disc diagrams that are free of pathologies provided that they do not contain siamese spiders. In Section 4 we introduce saturated prismatic presentations, which allow us to remove siamese spiders from disc diagrams, and we are able to use the previous tools to characterise geodesics in saturated prismatic presentations. In Section 5 we explore consequences of this characterisation (FFTP, shortlex automaticity, Tits-like rewriting system); lastly, Section 6 shows that several families of groups (trickle, TRAAGs and a generalisation of them) admit prismatic presentations. We observe that the definition of trickle groups in [4] allows for the existence of hexagonal moves, as it is possible to gather from Section 7 in the cited article.

2. PRISMATIC PRESENTATIONS

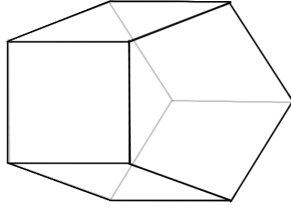
In this section we introduce our main object of study: prismatic presentations. The definition of this family of group presentations, despite its combinatorial appearance, carries an intuitive geometric interpretation which allows us to infer powerful properties of the groups they define.

A prismatic presentation is one where the set of relations can be split into two families: *squares*, which are relations of length four, and relations of varying length giving polygons which we called *heads*. These relations must satisfy some requirements: for example, one should not be able to glue a square to another 2-cell along a subword of length more than one (*small pieces* condition) and corners of cubes or prisms can be completed to full cubes or prisms (*cube completion* and *prism completion* conditions). This gives rise to two natural objects, *cubes* and *prisms*, which ensure that one can perform two operations on a disc diagram over a prismatic presentation: *hexagonal moves* and *prismatic moves*, which can be visualised in Fig. 6.

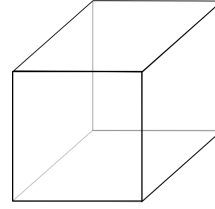
Let us formally define the basic objects that give geometric meaning to the definition of prismatic presentation that we will present afterwards.

Definition 2.1. A *prism* with base of length $n \in \mathbb{Z}_{\geq 2}$ is a CW -complex P with the following structure (see Fig. 4(A)):

- The 0-skeleton of P consists of $2n$ vertices.
- The 1-skeleton of P is built by attaching $3n$ 1-cells to the 0-skeleton as we describe next. Choose $V_1 \sqcup V_2$ a partition of the vertices of the 0-skeleton and enumerate them so that we have $V_1 = \{v_1, \dots, v_n\}$ and $V_2 = \{v_{n+1}, \dots, v_{2n}\}$. Attach $2(n-1)$ 1-cells joining vertices v_i and v_{i+1} for $i \in \{1, \dots, n-1\} \cup \{n+1, \dots, 2n-1\}$, as well as 1-cells joining v_n to v_1 , and v_{2n} to v_{n+1} . Attach the remaining n 1-cells so that they join v_i and v_{i+n} for $i \in \{1, \dots, n\}$.
- For every cycle of length four in the 1-skeleton that contains two vertices of V_1 and two vertices of V_2 , attach a square 2-cell, which will be referred to as a *lateral face*. For the cycles of length n with sets of vertices V_1 and V_2 attach respective 2-cells, which must be the interior of a polygon of n sides and will be called the *bases* of the prism. As a result, the 2-skeleton of P consists of $(2+n)$ 2-cells and P is simply connected.



(A) Prism with pentagonal base.



(B) Cube.

FIG. 4. Prisms and cubes.

Definition 2.2. A *cube* is a prism with base of length 4 (see Fig. 4(B)).

Definition 2.3. Let Σ be an alphabet. A Σ -prism (Σ -cube) is a prism (cube) whose 1-cells are oriented and labelled by elements of Σ . This induces a labelling by elements of $\Sigma \cup \Sigma^{-1}$ on the boundary of each two cell, which is well defined up to cyclic permutation (i.e. changing base point) and inversion (i.e. changing the orientation of the boundary).

Remark 2.4. Given a Σ -cube with three pairwise adjacent 2-cells $\sigma_1, \sigma_2, \sigma_3$ with the labelling indicated on the left, we can always label the remaining 2-cells of the Σ -cube τ_1, τ_2, τ_3 as indicated on the right:

$$\begin{aligned} \sigma_1 &= x_1 a_1 b_1 y_1^{-1}, & \tau_1 &= x_2 b_2 a_3 y_2^{-1}, \\ \sigma_2 &= y_1 a_2 b_2 z_1^{-1}, & \tau_2 &= y_2 b_3 a_1 z_2^{-1}, \\ \sigma_3 &= z_1 a_3 b_3 x_1^{-1}, & \tau_3 &= z_2 b_1 a_2 x_2^{-1}. \end{aligned}$$

Notice that the boundaries of the pairwise adjacent 2-cells $\sigma_1, \sigma_2, \sigma_3$ and τ_1, τ_2, τ_3 respectively are hexagones with the same labelling. See Fig. 5(A).

Remark 2.5. Given a more general Σ -prism, we can use the following standard notation for its labelling: the boundaries of the bases τ and ρ are labelled by

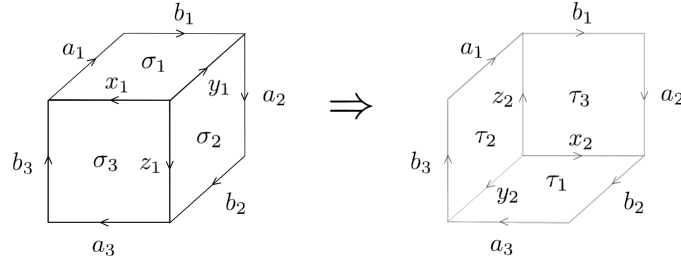
$$\tau = l_1 l_2 \dots l_n \quad \text{and} \quad \rho = r_1 r_2 \dots r_n,$$

and the lateral faces $\sigma_1, \dots, \sigma_n$ will have their boundaries labelled by

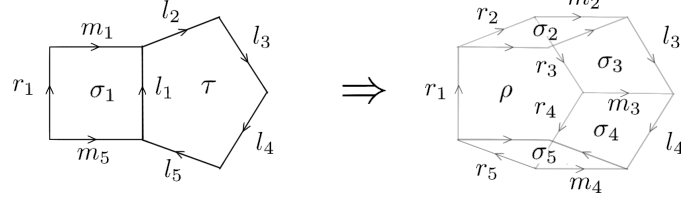
$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1} \quad \text{for } i = 1, \dots, n,$$

where $m_i, l_i, r_i \in \Sigma \cup \Sigma^{-1}$ under the convention $m_0 = m_n$. See Fig. 5(B).

We now may give the definition of the type of presentation we will be working with for the rest of the paper. The reader should note that the notation in this definition has been chosen so as to match the one in Remarks 2.4 and 2.5.



(A) Condition **P3**: cube completion. σ_i, τ_i denote 2-cells, a_i, b_i, x_i, y_i, z_i denote 1-cells.



(B) Condition **P4**: prism completion. σ_i, τ, ρ denote 2-cells, m_i, l_i, r_i denote 1-cells.

FIG. 5. Conditions for prismatic presentation following notation in Definition 2.6, see Fig. 1 for an unlabelled version.

Definition 2.6 (Prismatic presentation). Let $\langle \Sigma \mid R \rangle$ be a group presentation with generating set Σ and a set of cyclically reduced relations R for a group G . We say that this presentation is prismatic if the set of relations admits a partition $R = S \sqcup H$ satisfying the following conditions:

- P1** (Squares and heads) Every $\sigma \in S$ is a word of length 4. The cells in the corresponding Cayley complex associated to these relations will be called squares, and cells associated to H will be called heads (note that heads are also allowed to have length four).
- P2** (Small pieces) Given $\sigma \in S$ and $\tau \in S \sqcup H$, if a cyclic permutation of σ or σ^{-1} shares a subword of length 2 with τ then $\tau = \sigma$.
- P3** (Cube completion) For every triple $\sigma_1, \sigma_2, \sigma_3$ of cyclic permutations of relations in $S \cup S^{-1}$ that can be factorized as

$$\begin{aligned} \sigma_1 &= x_1 a_1 b_1 y_1^{-1} \\ \sigma_2 &= y_1 a_2 b_2 z_1^{-1} \\ \sigma_3 &= z_1 a_3 b_3 x_1^{-1}, \end{aligned}$$

where $a_i, b_i, c_i, x_1, y_1, z_1 \in \Sigma \cup \Sigma^{-1}$ and $a_1 b_1 a_2 b_2 a_3 b_3$ is a cyclically reduced word, there exists another triple τ_1, τ_2, τ_3 of cyclic permutations of relations

$S \cup S^{-1}$ such that

$$\begin{aligned}\tau_1 &= x_2 b_2 a_3 y_2^{-1} \\ \tau_2 &= y_2 b_3 a_1 z_2^{-1} \\ \tau_3 &= z_2 b_1 a_2 x_2^{-1}.\end{aligned}$$

Note that these six relations provide the labelling of a Σ -cube. See Fig. 5(A).

P4 (Prism completion) For every σ_1 that is a cyclic permutation of a relation in $S \cup S^{-1}$ and τ a cyclic permutation of a relation in $H \cup H^{-1}$, where

$$\begin{aligned}\sigma_1 &= l_1 m_1^{-1} r_1^{-1} m_n, \\ \tau &= l_1 l_2 \dots l_n,\end{aligned}$$

with $l_i, m_i, r_i \in \Sigma \cup \Sigma^{-1}$, there is a relation ρ that is a cyclic permutation of an element in $H \cup H^{-1}$ and there are relations $\sigma_2, \dots, \sigma_n$ that are cyclic permutations of elements in $S \cup S^{-1}$ of the form

$$\begin{aligned}\rho &= r_1 r_2 \dots r_n, \\ \sigma_i &= l_i m_i^{-1} r_i^{-1} m_{i-1},\end{aligned}$$

with $r_i, l_i, m_i \in \Sigma \cup \Sigma^{-1}$. Note that these $(n+2)$ relations provide the labelling of a Σ -prism. See Fig. 5(B).

A few remarks are worth making for the extreme cases where S or H are empty.

Let G be a group acting on a CAT(0) cube complex X with a free and transitive action on the 0-skeleton. Then G naturally admits a prismatic presentation with no heads. Indeed, take Σ to be a set of elements that move a fixed vertex v_0 to its neighbors. Then the Cayley graph of G with respect to Σ is the 1-skeleton of X , and we can take S to be the labels of the 2-cubes in X . So, the case of prismatic presentations with no heads should be regarded as a generalisation of groups acting on CAT(0) cube complexes with an orbit on the 0-skeleton. In this sense, our Theorem C can be regarded as a generalisation of a theorem of Noskov [27].

Note that any (cyclically reduced) group presentation $\langle \Sigma \mid R \rangle$ is prismatic taking $H = R$. We will establish conditions on the set H later for specific applications. For a head $\tau \in H$, consider the minimal subset $H(\tau) \subseteq H$ with the following properties: (i) $\tau \in H(\tau)$ and (ii) for any other head $\tau' \in H$ sharing a letter with some head in $H(\tau)$, one has that $\tau' \in H(\tau)$. Let us set $\Sigma_{H(\tau)}$ to be the set of letters appearing in some head of $H(\tau)$. Note that for $\tau, \tau' \in H$ either $H(\tau) = H(\tau')$, or $H(\tau)$ and $H(\tau')$, as well as $\Sigma_{H(\tau)}$ and $\Sigma_{H(\tau')}$, are disjoint. We call $\langle \Sigma_{H(\tau)} \mid H(\tau) \rangle$ a *spiderhead presentation* of $\langle \Sigma \mid R \rangle$.

Definition 2.7. We will mostly consider two types of spiderhead presentations:

- (1) Cyclic spiderheads: presentations of the form $\langle a \mid a^n \rangle$.
- (2) Triangular spiderheads: given a group G consider the presentations with generating set $G - \{1_G\}$ and as relations all the words of length 3.

We emphasise that our conditions are on the spiderhead presentation and not on the group they define. For instance if a spiderhead presentation is of the form $\langle a \mid a^3, a^6 \rangle$, this is not a cyclic spiderhead (even if it defines a cyclic group).

The definition of prismatic presentation does not guarantee the existence of cubes and prisms in the Cayley complex, but the existence of combinatorial maps from cubes and prisms into it. This will allow to perform hexagonal and prismatic moves in the van Kampen diagrams (See Figure 3). Moreover, the fact that these cubes and prisms embed injectively into the Cayley complex will be a consequence of

the solution to the word problem obtained in Theorem 4.23 for presentations with certain spiderhead presentations.

2.1. Examples of prismatic presentations. We are going to give several interesting examples of prismatic presentations, but we will defer the proofs to the last sections of the paper as we prefer to focus first on the power of disc diagrams over prismatic presentations.

Before continuing we fix some standard notation that will be used throughout the paper.

Notation 2.8. The length of a word w will be denoted by $|w|$. Given a generating set Σ of a group G and two words $w, v \in (\Sigma \cup \Sigma^{-1})^*$ we write $w =_G v$ if they represent the same element in G . The length of the shortest $v \in (\Sigma \cup \Sigma^{-1})^*$ such that $v =_G w$ is denoted by $|w|_G$. We say w is geodesic if $|w| = |w|_G$.

Trickle groups. The initial motivation of this paper was obtaining geometric tools to work with trickle groups. These groups were introduced by Bellingeri, Godelle and Paris in [4] and the family of groups they constitute contains well-known classes as right-angled Artin/Coxeter groups, cactus groups and even Thompson's group F , which admits an (infinite) trickle presentation. In that sense, one of our main theorems is

Theorem 2.9. *Any trickle presentation $Tr(\Gamma)$ is a prismatic presentation with cyclic spiderheads.*

A trickle presentation is given by a quadruple $(\Gamma, \leq, \mu, (\varphi_x)_{x \in V(\Gamma)})$, where

- Γ is a simplicial graph,
- $\leq$ is a (partial) order on $V(\Gamma)$,
- μ is a labelling $\mu: V(\Gamma) \rightarrow \mathbb{N}_{\geq 2} \cup \{\infty\}$ of the vertices,
- $\varphi_x: \text{Star}_\Gamma(x) \rightarrow \text{Star}_\Gamma(x)$ is a graph automorphism of $\text{Star}_\Gamma(x)$ for all $x \in V(\Gamma)$.

This quadruple has to satisfy seven axioms (see Section 6.1 for details) and, using z_t to denote $\varphi_t(z)$ for $\{z, t\} \in E(\Gamma)$, the group presentation obtained from this quadruple is as follows:

$$Tr(\Gamma) = \langle V(\Gamma) \mid x^{\mu(x)} = 1 \text{ for } x \in V(\Gamma) \text{ s.t. } \mu(x) \neq \infty, y_x x = x_y y \text{ for } \{x, y\} \in E(\Gamma) \rangle.$$

Torsion T-RAAGs. One of the advantages of dealing with prismatic presentations is that they encompass presentations that are very similar to trickle presentations but produce non-trickle groups (see Remark 6.15 below). We introduce first some notation.

Definition 2.10 (Decorated Mixed graph). A *mixed graph* $\Gamma = (V, E, D, \iota, \tau)$ consists of an *underlying simplicial graph* (V, E) , a set of *directed edges* $D \subseteq E$, and maps $\iota, \tau: D \rightarrow V$ giving the *initial* and *terminal* vertex of edges. For each $e = \{x, y\} \in D$, we have $\iota(e), \tau(e) \in \{x, y\}$ and $\iota(e) \neq \tau(e)$. A *decorated mixed graph* $\Gamma = (V, E, D, \iota, \tau, \mu)$ is a mixed graph $\Gamma = (V, E, D, \iota, \tau)$ equipped with a labelling map $\mu: V \rightarrow \mathbb{N} \cup \{\infty\}$.

Notation 2.11. Let Γ be a decorated mixed graph $(V, E, D, \iota, \tau, \mu)$.

- (i) If $e = \{a, b\} \in E \setminus D$, we use the notation $[a, b]$, or equivalently $[b, a]$, to refer to the edge e .
- (ii) If $e = \{a, b\} \in D$, we use the notation $[\iota(e), \tau(e)]$ to refer to the edge e .

Definition 2.12 (torsion T-RAAG). Let $\Gamma = (V, E, D, \iota, \tau, \mu)$ be a decorated mixed graph. The *torsion twisted right-angled Artin group* based on Γ is given by the quotient

$$T_{\text{tor}}(\Gamma) = \left\langle V \left| \begin{array}{l} ab = ba \text{ if } [a, b] \in E - D, \\ aba = b \text{ if } [a, b] \in D, \\ a^{\mu(a)} = 1 \text{ for } a \in V \text{ with } \mu(a) < \infty \end{array} \right. \right\rangle$$

Theorem 2.13. Let $\Gamma = (V, E, D, \iota, \tau, \mu)$ be a decorated mixed graph where the labelling of the vertices satisfies that, for every $[u, v] \in D$, $\mu(v)$ is even or $\mu(v) = \infty$. Then the T-RAAG presentation $T_{\text{tor}}(\Gamma)$ is a prismatic presentation with cyclic spiderheads.

Graph products. The definition of prismatic presentation naturally guarantees closure under taking *graph products*. Consider $\Gamma = (V\Gamma, E\Gamma)$ a simplicial graph and a collection $\{G_v = \langle \Sigma_v \mid S_v \cup H_v \rangle\}$ of prismatic presentations associated to the vertices in $V\Gamma$. Consider the generating set $\Sigma = \sqcup_{v \in V\Gamma} \Sigma_v$, the set of heads $H = \sqcup_{v \in V\Gamma} H_v$ and the set of squares

$$S = (\sqcup_{v \in V\Gamma} S_v) \cup \{ab = ba \text{ for } a \in \Sigma_u, b \in \Sigma_w, \{u, w\} \in E\Gamma\}.$$

The presentation $\langle \Sigma \mid H \cup S \rangle$ defines the graph product $\mathcal{G} = \mathcal{G}(\Gamma, \{G_v\}_{v \in \Gamma})$ of the groups G_v associated to the graph Γ .

Proposition 2.14. Let $\langle \Sigma \mid H \cup S \rangle$ be the presentation, as described above, of the graph product $\mathcal{G}(\Gamma, \{G_v\}_{v \in \Gamma})$ coming from a simplicial graph $\Gamma = (V\Gamma, E\Gamma)$ and a collection of prismatic presentations $\{G_v = \langle \Sigma_v \mid S_v \cup H_v \rangle\}$. The presentation $\langle \Sigma \mid H \cup S \rangle$ is a prismatic presentation.

Moreover spiderhead presentations of $\langle \Sigma \mid H \cup S \rangle$ coincide with spiderhead presentations of the vertex groups.

Tits' algorithm to solve the word problem in Coxeter groups is given by a rewriting system defined by two families of rules $M_I \sqcup M_{II}$: the rules in M_I reduce the length of the a word while the ones in M_{II} do not and are reversible.

This algorithm has been generalised to other families of groups, including Trickle groups [4, 11], among others. We note that Green's normal form for graph products [15] is also a Tits-style rewriting system where the generating set is infinite and consists of the union of the elements of the vertex sets. We provide a common generalisation in the framework of prismatic presentations.

Definition 2.15 (Tits-like rewriting system). Let $\langle \Sigma \mid H \cup S \rangle$ be a prismatic presentation. We define the following rewriting system for the words over Σ . Letting ε denote the empty word, set $M = M_I \cup M_{II}$, where

$$\begin{aligned} M_I &= \{u \rightarrow v \mid |v| < |u|, uv^{-1} \in H \text{ (up to cyclic permutation/inverse)}\} \cup \\ &\quad \{x^\tau x^{-\tau} \rightarrow \varepsilon \mid x \in \Sigma, \tau \in \{1, -1\}\}, \\ M_{II} &= \{u \rightarrow v \mid |v| = |u|, uv^{-1} \in S \sqcup H \text{ (up to cyclic permutation/inverse)}\}. \end{aligned}$$

For $X \in \{M_I, M_{II}, M\}$, we write $u \xrightarrow{X} v$ if a word v is obtained from a word u in one X -rewriting step, and $u \xrightarrow{X^*} v$ if v is obtained from u in zero or more X -rewriting steps. We say that a word is M -reduced if it cannot be shortened by performing M -rewriting steps.

We have:

Theorem 2.16. *Let $\langle \Sigma \mid H \cup S \rangle$ be a saturated prismatic presentation (see Definition 4.2) and $M = M_I \cup M_{II}$ as above. Then*

- (1) *A word w over Σ is geodesic if and only if it is M -reduced.*
- (2) *Two geodesic words u, v over Σ represent the same group element if and only if $u \xrightarrow{M_{II}^*} v$.*

Our proof is based on disc diagrams and it is of a different nature than [15, 4, 11]. The proof is similar in style to [22, §4].

3. CORRIDORS, SPIDERS AND PATHOLOGIES

3.1. Disc diagrams over prismatic presentations. We are going to translate the conditions **P2**, **P3** and **P4** of the definition of a prismatic presentation to a geometric point of view. Our main tool for understanding prismatic presentations geometrically will be the language of *disc diagrams*.

Definition 3.1. A map $f: X \rightarrow Y$ between 2-dimensional CW-complexes is *combinatorial* if it maps open cells homeomorphically to open cells. A complex is combinatorial if all attaching maps are combinatorial.

Definition 3.2 (Disc diagram). A *disc diagram* D is a compact contractible combinatorial 2-complex, together with an embedding $D \hookrightarrow \mathbb{E}^2$. Let X be a 2-complex. A *disc diagram in X* is a disc diagram $D \hookrightarrow \mathbb{E}^2$ together with a combinatorial map $D \rightarrow X$. The *area* of a disc diagram D is the number of 2-cells in D .

In our setup, we will work with disc diagrams in a specific complex:

Definition 3.3 (Presentation complex). Given a group presentation $P = \langle S \mid R \rangle$, its *standard presentation complex* has a single 0-cell v , a 1-cell attached (and labelled) by each generator in the presentation, and a polygonal 2-cell for each relator in the presentation which is attached along the corresponding 1-cells that label the relator. The *Cayley complex* X_P of the presentation is the universal cover of the standard presentation complex.

Remark 3.4. In the context of a CW-complex, we will use the term *path* to refer to a combinatorial path in its 1-skeleton. The 1-cells of a disc diagram D in the Cayley complex X_P inherit a natural labelling from the one in X_P , and so a path in D carries a label determined by the labels of its edges.

Definition 3.5. A *boundary cycle* of a disc diagram D in X_P is a closed path that travels along the whole boundary of the disc diagram respecting the planar embedding of the diagram. A *boundary label* of D is a word on the generators of the presentation P which labels a boundary cycle of D .

Notice that the map from a disc diagram D to the Cayley complex X_P is not necessarily an immersion:

Definition 3.6. A disc diagram D in X_P is *reduced* if it does not contain two 2-cells B_1, B_2 sharing a 1-cell e in $\partial B_1 \cap \partial B_2$ in such a way that the label of the boundary of both cells starting at the shared one-cell coincides (that is, ∂B_1 and ∂B_2 map to the same closed path in X_P).

Remark 3.7. It is possible to replace a non reduced disc diagram D in X_P with another disc diagram D' in X_P which has the same boundary cycle but smaller

area: just remove e and the interior of B_1 and B_2 and then glue $\partial B_1 - e$ with $\partial B_2 - e$.

The boundary label of a disc diagram must represent the identity in the group; a converse statement is also true (see [6] for a proof).

Lemma 3.8 (Van Kampen's Lemma). *Given a presentation for a group and a word w over the generators that represents the identity element, there is a (reduced) disc diagram in the corresponding Cayley complex whose boundary cycle is labelled by w .*

We are ready to translate the conditions **P2**, **P3** and **P4** of the definition of a prismatic presentation as moves in disc diagrams:

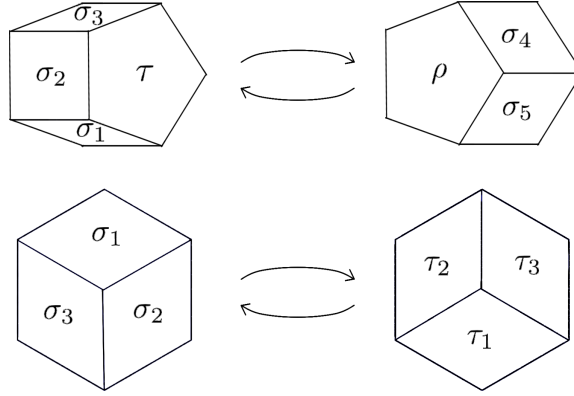


FIG. 6. Prismatic moves (above) and hexagonal moves (below).

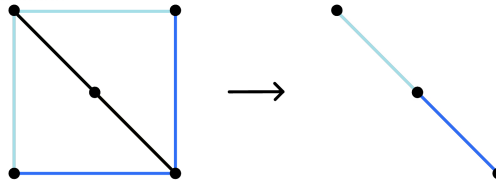


FIG. 7. Collapse of two squares.

Definition 3.9 (Collapse). Let D be a disc diagram over a prismatic presentation. Note that if D contains two square 2-cells induced by the same relation (up to cyclic conjugation), and these two squares are glued along two consecutive edges, we can retessellate D to a diagram D' so that these two squares are replaced by two edges joined by a vertex. Similarly, if two square 2-cells induced by the same relation (up to cyclic permutation) are glued along three consecutive edges, we can replace the two squares by a single edge. This operation is what we call a *collapse*. Note that $a(D) = a(D') + 2$.

Definition 3.10. (Half-cube and hexagonal move) Consider a disc diagram D over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$. A half-cube (or tessellated hexagon) in D is a subdiagram of D induced by three pairwise adjacent 2-cells that are associated to relations $\sigma_1, \sigma_2, \sigma_3 \in S$ as in **P3**. Condition **P3** ensures that, whenever D contains a half-cube induced by relations $\sigma_1, \sigma_2, \sigma_3$ as a subdiagram, this subdiagram can be retessellated using the other half of the cube that σ_1, σ_2 and σ_3 induce, given by the corresponding relations τ_1, τ_2, τ_3 . This retessellation is what we call a *hexagonal move* (see Fig. 6).

Definition 3.11. (Half-prism and prismatic move) Consider a disc diagram D over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$. A half-prism in D is a subdiagram consisting of one 2-cell associated to a relation $\tau \in H$ and several 2-cells associated to relations $\sigma_i, \sigma_{i+1}, \dots, \sigma_{i+k}$ as in **P4**. Condition **P4** ensures that, whenever D contains a n -gonal half-prism induced by relations $\tau \in H$ and $\sigma_i, \dots, \sigma_{i+k} \in S$ for some $k \geq 1$ as a subdiagram, this subdiagram can be retessellated using the remaining 2-cells of the prism that the half-prism induces, given by the corresponding relations $\rho \in H$ and $\sigma_j \in S$, with $j \notin [i, i+k]$. This retessellation is what we call a *prismatic move* (see Fig. 6).

Remark 3.12. If a disc diagram over a prismatic presentation contains a half-prism with one base, its retessellation contains the other base that has been ‘pushed away’ (see Fig. 6). This will be used to choose disc diagrams where the placement of the bases is convenient for our purposes.

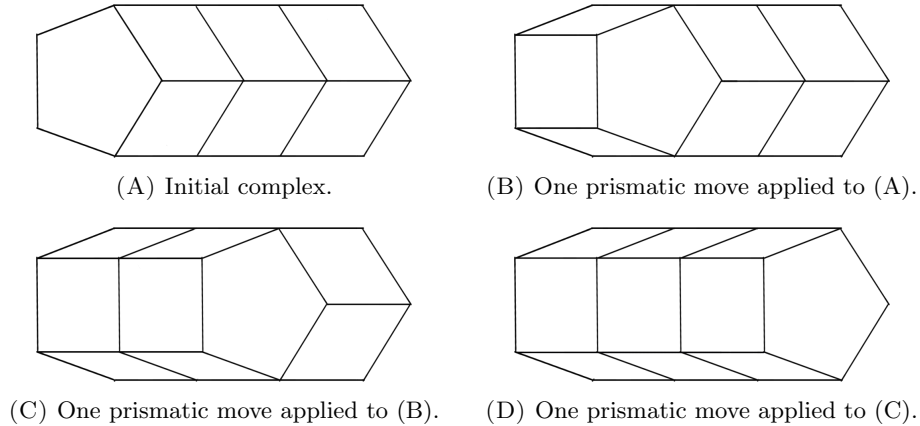


FIG. 8. Prismatic moves used sequentially to deal with longer prisms, where first and last complex are understood as prisms of height three.

Prisms have been defined to have height equal to 1 (that is, the distance between the two bases is 1), but we will often apply prismatic moves sequentially to work with disc diagrams that contain ‘prisms’ of height greater than 1 as shown in Fig. 8. This is strictly a sequence of applications of prismatic moves but, for short, we will refer to this sequence simply as a prismatic move.

Definition 3.13. Two disc diagrams D and D' over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$ are *M-equivalent* if they have the same boundary label and there is a

sequence of collapses/uncollapses, hexagonal and prismatic moves that takes D to D' .

3.2. Corridors, spiders and pathologies. In this subsection we are going to define some subdiagrams called corridors and spiders. When we choose disc diagrams over prismatic presentations, there are certain configurations that corridors and spiders can exhibit that we wish to avoid. We call them *pathological* configurations (Definition 3.25).

Definition 3.14 (Corridor). Let D be a disc diagram over a prismatic presentation. We say that two edges e and e' of D are *parallel* if they are opposite sides of a square 2-cell in D . The transitive closure of the relation that identifies parallel edges gives an equivalence relation defined on the set of edges of D . Given an edge e of D , the *corridor induced by e* is the subcomplex induced by the 2-cells of D which contain two parallel edges in this equivalence class.

Definition 3.15 (Intersecting/perpendicular). We say that two distinct corridors C_1, C_2 (not necessarily distinct) intersect at a square s in $C_1 \cap C_2$ (or, equivalently, are perpendicular at a square s) if there are edges e, f inducing C_1 and C_2 respectively that are adjacent sides of the square s .

Definition 3.16 (Length of a corridor). Consider a disc diagram D and a corridor C , the *length* of C in D is given by the number of 1-cells in the equivalence class of C minus one. A zero-length corridor will also be referred to as a *trivial* corridor.

Definition 3.17 (Spider). Given a disc diagram D over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$, a *head* is a polygonal 2-cell in D arising from a relation in H . The (possibly trivial) corridors in D induced by the edges of a head are called *legs*. The subdiagram of D given by the union of a head and its associated legs is what we call a *spider* diagram.

Lemma 3.18. *Let D be a disc diagram over a prismatic presentation and D' be an M -equivalent diagram. Let e, f be edges in $\partial D = \partial D'$, then e and f induce the same corridor in D if and only if they induce the same corridor in D' .*

Proof. It is enough to check the statement when we apply to D a single move. And this is clear for each type of move from Figures 6 and 7. $\square$

We introduce now the four *corridor pathological configurations* and we illustrate each of them in Fig. 9. Notice that the corridors involved in these pathologies might be legs of spiders.

Definition 3.19 (Circular corridor). Let C be a corridor. If there is a sequence of edges $(e_i)_{i=1}^n$, with $n \geq 2$, defining the corridor such that e_i is parallel to e_{i+1} and $e_1 = e_n$, we say that C is a *circular* corridor.

Identifying each square in the corridor with the space $[-1, 1] \times [-1, 1]$, where the points $(0, \pm 1)$ correspond to the midpoints of the edges defining the corridor, we define the *fibre* of the circular corridor as the simple, closed curve given by the union of the segments $\{0\} \times [-1, 1]$ for each square in the corridor.

Definition 3.20 (Self-osculating corridor). Given a corridor C , if there is a sequence of edges $(e_i)_{i=1}^n$ defining the corridor such that e_i is parallel to e_{i+1} and e_1, e_n share exactly one vertex, we say that the corridor C *self-osculates*.

LAM: andrew points out repetition between this def of intersect at a square, and the def of perpendicular later on. Remove def later on and add it here? PLL: he movido aquí lo de perpendicular y he traído aquí la definición para que nos sirva también para intersecting legs (si hacía la separación más abajo parecía que esto fuese una patología).

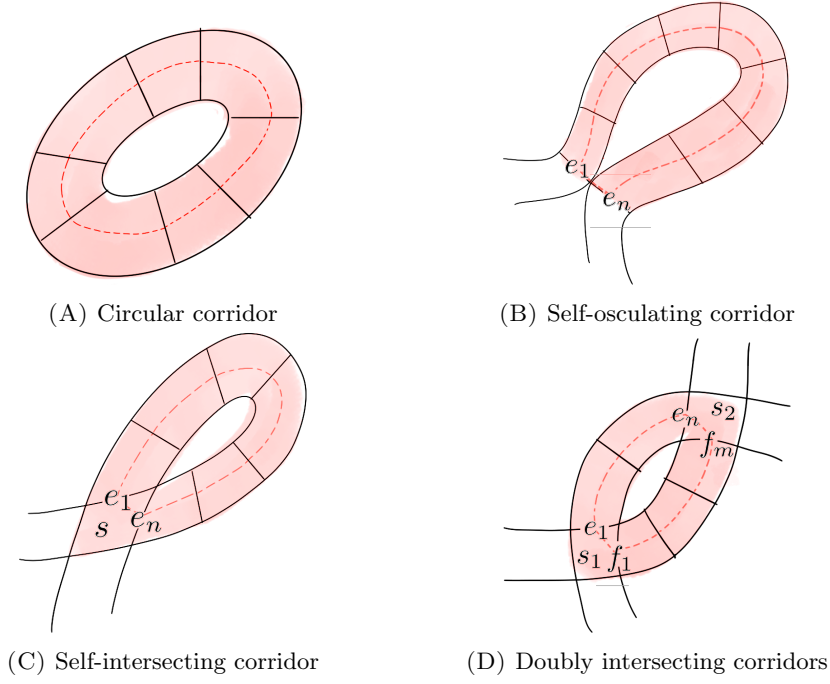


FIG. 9. In each image the dotted line shows a fibre and the shaded cells show the corresponding pathology P .

If the sequence $(e_i)_{i=1}^n$ is minimal, in the sense that there are no edges e_i, e_j with $1 \leq i < j < n$ and e_i, e_j sharing exactly one vertex, we define its fibre as follows. With the notation above, take the segments $\{0\} \times [-1, 1]$ for the squares induced by the pairs (e_i, e_{i+1}) . The union of these segments, together with the straight line joining the midpoint of e_1 with the midpoint of e_n , is a simple, closed curve which we call the *fibre* of the self-osculation.

Definition 3.21 (Self-intersecting corridor). A corridor C *self-intersects at a square* s in C if there is a sequence of edges $(e_i)_{i=1}^n$ defining the corridor such that e_i is parallel to e_{i+1} and e_1, e_n are adjacent edges of the square s .

If the sequence is minimal, in the sense that there is no shorter subsequence exhibiting this self-intersecting behaviour, we consider again the segments $\{0\} \times [-1, 1]$ for the squares induced by the pairs (e_i, e_{i+1}) in the sequence. The union of these segments with the straight line in s that joins the midpoint of e_1 to the midpoint of e_n is a simple, closed curve called the *fibre* of the self-intersection.

Definition 3.22 (Doubly intersecting corridors). We say that there is a *double intersection* between two corridors if they intersect at two different squares. In this case, we can define sequences $(e_i)_{i=1}^n$ and $(f_i)_{i=1}^m$ of edges inducing C_1 and C_2 respectively so that the pairs (e_i, e_{i+1}) and (f_i, f_{i+1}) give parallel edges, e_1, f_1 are consecutive edges in a square s_1 in $C_1 \cap C_2$, e_n, f_m are consecutive edges in a square s_2 in $C_1 \cap C_2$.

If these sequences are minimal in the sense that no subsequences exhibit a double intersection, we define its fibre as follows. We take once more the segments $\{0\} \times$

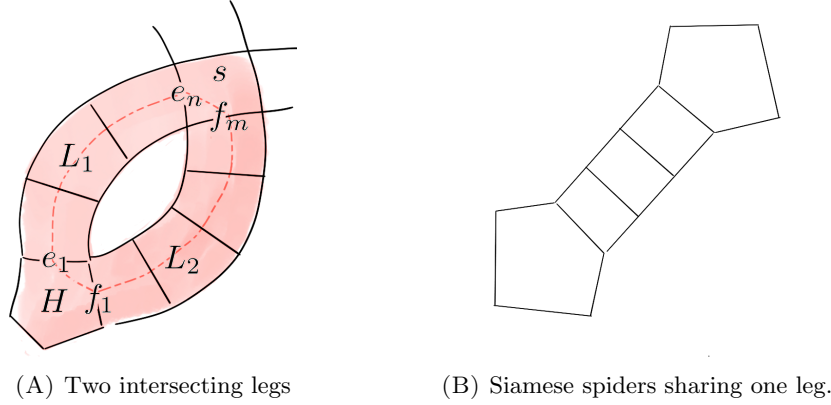


FIG. 10

$[-1, 1]$ for the squares induced by the pairs (e_i, e_{i+1}) and by the pairs (f_i, f_{i+1}) , and we take the union of these segments with the straight line in s_1 joining the midpoints of e_1 and e_n , and the straight line in s_2 joining the midpoints of e_n and f_m . This union yields what we call the *fibre* of the double intersection, which is again a simple, closed curve.

We also have pathologies involving a head, and we call them *spider pathological configurations*. They can be of two types:

Definition 3.23 (Intersecting legs). A spider has a pair of intersecting legs if two of its legs L_1 and L_2 (corridors) intersect at a square s . If the sequences of parallel edges $\{e_i\}^n \subset L_1$ and $\{f_i\}^m \subset L_2$, where e_1, f_1 are edges in the head and e_n, f_m are consecutive sides of s , are chosen to be minimal in the sense that no pair $(e_i, f_j) \neq (e_n, f_m)$ exhibits an intersection of L_1 and L_2 , we define the *fibre* of the leg intersection as follows. Consider the segments $\{0\} \times [-1, 1]$ for all squares with opposite sides (e_i, e_{i+1}) and (f_i, f_{i+1}) . Taking the union of these segments together with a straight line in s joining the midpoint of e_n and f_m and a straight line in the head H joining the midpoint of e_1 to the midpoint of f_1 , we obtain a simple, closed curve which we call fibre of the intersection.

Definition 3.24 (Circular leg). Suppose that e and f are two edges in the boundary of a head that induce the same corridor L . Then we say that L is a circular leg.

Identifying each square in the leg L with the space $[-1, 1] \times [-1, 1]$, where the points $(0, \pm 1)$ correspond to the midpoints of the edges defining the corridor, we define the *fibre* of the circular leg as the simple, closed curve given by the union of the segments $\{0\} \times [-1, 1]$ for each square in the leg together with a segment in the head joining the midpoints of e and f .

Definition 3.25 (Pathology, internal area). In the phenomena defined above, the 2-complex spanned by the 2-cells whose interiors intersect the fibre is what we call a *pathology* P . We will say that P is a corridor pathology if it comes from a corridor pathological configuration and we will say that P is a spider pathology if it comes from a spider pathological configuration. Given a pathology P in a disc diagram D , Jordan's theorem ensures that its fibre separates the diagram in an interior and an exterior region. The 2-complex given by the union of the interior region with

LAM: añadir imagen circular y mover siamesa

LAM: andrew points out repetition with intersecting corridors. he says maybe this one can be shortened(?) **PLL:** hecho

the pathology P is what we call the *diagram enclosed by the pathology*, denoted $\bar{P}$. The *internal area* of the pathology $a(P)$ is given by the number of 2-cells in $\bar{P} - P$.

We gather in the following definitions some important concepts that will appear in the theorems of this section.

Definition 3.26 (Side of a corridor). Let D be a squared disc diagram. Let C be the corridor induced by an edge e in D . Since D can be embedded on the plane, it cannot contain a non-orientable surface. Therefore, if we remove the (interior of the) 1-cells of C that are in the equivalence class of C and all of the 2-cells, we obtain two one-dimensional complexes that we will call the *sides of the corridor*. We will mostly refer to the sides of corridors which are not pathological, in particular, these corridors will have exactly two sides.

We finalise this subsection with the concept of arc and spider arc that will be important later, as our main technical result in the next section guarantees the existence of certain boundary arcs that will allow to "reduce" our diagrams.

Definition 3.27 (Arc). Let D be a disc diagram over a prismatic presentation. An *arc* in D is a non-pathological corridor whose equivalence class contains two different edges e and e' of ∂D . The edges e and e' are called the *extremes* of the arc.

Definition 3.28 (Spider arc). Let D be a disc diagram over a prismatic presentation. Given a non-pathological spider in D , a *spider arc* is given by the union of its head H together with two of its legs. The two edges where these legs intersect ∂D are called the *extremes* of the spider arc. Given a spider arc with extremes e , e' , consider its 1-skeleton and remove from it the edges in the equivalence classes of e and e' . Since D embeds on the plane we obtain two 1-dimensional complexes that we call the *sides* of the spider arc.

Definition 3.29 (Outermost arc, outermost boundary arc). Let D be diagram over a prismatic presentation. We say that an arc A with extremes e and e' is *outermost* with respect to one of the components of $\partial D - \{e, e'\}$ if all the corridors induced by the edges in this component intersect the arc A . We say that an arc A with extremes e and e' is an *outermost boundary arc* if one of its sides is contained in ∂D and it is outermost with respect to both components of $\partial D - \{e, e'\}$.

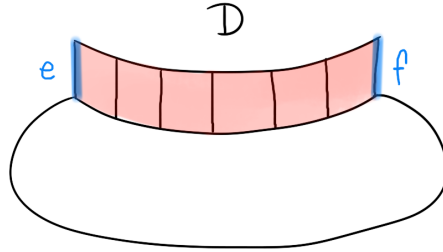


FIG. 11. Diagram with an arc at its boundary.

The analogous definition for outermost spider arc will be discussed later (Definition 4.11).

LAM: Montse had another ref for this, ol'shanskii? we could as her. Hagen attributes the arguments to someone called Casson (typo of Cannon?). **PLL:** En un paper de Mark sapir se citan muchos papers de Ol'shanskii pero no sé cuál exactamente será el que nos interesa como referencia

3.3. Existence of squared diagrams free of pathologies. The results in this subsection and the next will ensure the existence of diagrams with no pathologies. Similar ideas can be found in the literature (see [17, 31, 29, 16]). We will first deal with the case of diagrams with no heads.

Definition 3.30 (Squared diagram). Let $\langle \Sigma | S \sqcup H \rangle$ be a prismatic presentation. We say that a disc diagram D over this presentation is *squared* if all its 2-cells are squares (that is, they are associated to relations in S).

In this subsection we show that every squared diagram D over a prismatic presentation is M -equivalent to a squared diagram D' without pathologies satisfying that two edges in $\partial D = \partial D'$ induce a corridor in D if and only if the corresponding edges in $\partial D'$ induce a corridor in D' . See Theorem 3.35.

The following is an auxiliary lemma that will be used in Theorem 3.32.

Lemma 3.31. *Let D be a squared disc diagram over a prismatic presentation. Assume D contains no pathologies. If D' is a disc diagram obtained from D by applying a hexagonal move, then D' contains no pathologies.*

Proof. Let D' be the resulting disc diagram after applying a hexagonal move to D . We will see that any pathology in D' must come from a pathology in D . Let H be a tessellated hexagon in D and let H' be its alternative tessellation in D' . Since $\partial H = \partial H'$ and $D \setminus H$ remains the same after retessellating, for any edge e of $\partial H \cup (D - H)$ in D , there is a natural corresponding edge e' of $\partial H' \cup (D' - H')$ in D' . We may think of the corridor $C_{e'}$ (induced by e') as being the corridor C_e (induced by e) after the retessellation. Notice that only the corridors induced by edges in ∂H might change after the hexagonal move. Also, observe that two opposite edges in $\partial H'$ induce the same corridor. Let e be an edge of ∂H and suppose that $C_{e'}$ is involved in some pathology.

- If $C_{e'}$ is a circular corridor so must be C_e , since these two corridors are determined by the same edges in $\partial H = \partial H'$ and D, D' only differ in the hexagon.
- If $C_{e'}$ is a self-intersecting corridor, and the self-intersection point occurs out of H' , the self-intersection occurs in D for the same reason as before: D and D' only differ in the hexagon and the pairs of edges in $\partial H = \partial H'$ determining corridors have not changed. If the self-intersection point occurs in H' it is immediate to see that there was already a self-intersecting point in H for C_e .
- If $C_{e'}$ is a self-osculating corridor and the self-osculating point occurs outside of H' , it is immediate to see that the self-osculation was already exhibited by C_e in H . If the self-osculating point appears in H' , note that it cannot be the interior vertex of H' because the self-osculation vertex must belong, by definition, to at least 4 squares. Also, the self-osculation point cannot be the one vertex v of the hexagon which does not belong to $C_{e'}$. Suppose the self-osculation vertex is the one opposite v in the hexagon. If we undo the hexagonal move, we now have a square of H adjacent to the vertex corresponding to the self-osculating point in D' . Take the edge of $\partial H = \partial H'$ that lies inside the interior region of the pathology in D' . The corridor induced by this edge crosses C_e twice: inside and outside H . Now suppose that the self-osculation point is one of the other vertices. If the intersection of the hexagon with the interior of the self-osculation

consists only of this vertex, undoing the hexagonal move gives again a self-osculation. If the interior region of the self-osculation contains v , undoing the hexagonal move gives again a self-osculation. If the intersection of the hexagon with the interior region of the self-osculation consists of the two edges in $\partial H'$ that are adjacent to the vertex opposite v , undoing the hexagonal move gives a double intersection between the C_e and the corridor induced by one of these two edges.

- If $C_{e'}$ exhibits a double intersection with none of the two intersection squares belonging to H' , it is clear from the fact that the hexagonal move only affects H' that the same two corridors exhibited a double intersection with C_e in D . If one of the intersection squares belongs to H' , call $C_{f'}$ the other corridor involved in the intersection and call C_f the corresponding corridor in D . Undoing the hexagonal move on H' , it is also clear that one of the squares in H is an intersection square for two corridors defined by C_e and C_f , so the double intersection was exhibited by D .

□

To clarify the terminology in the next theorem, we note that when we say that two squares in a disc diagram are adjacent, we mean that they share at least one of their sides. In particular, two squares that share only one vertex are not adjacent.

Theorem 3.32. *Let $D = B \cup R$ be a squared disc diagram over a prismatic presentation as in Fig. 12. Assume that D is free of pathologies, B is an outermost boundary arc and R has positive area. Then there are adjacent squares t and t' in B and a square s in R adjacent to both of them. In particular, one can perform a hexagonal move that yields a new diagram $D' \cup s'$ where:*

- (1) $D' = B' \cup R'$ where B' is an outermost boundary arc of D' ,
- (2) The area of R' is smaller than the area of R ,
- (3) s' is a square 2-cell which does not intersect R' ,
- (4) $\partial D = \partial(D' \cup s')$,
- (5) The arcs B and B' are induced by the same two edges of ∂D , and
- (6) $D' \cup s'$ has no pathologies.

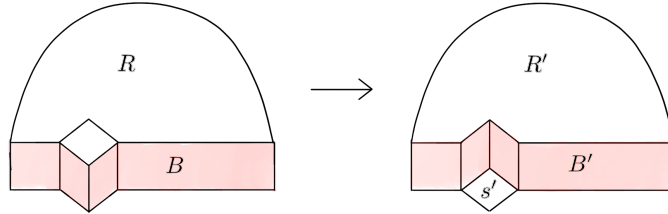


FIG. 12. Setup of Theorem 3.32

Proof. Since B is an outermost boundary arc and D contains no pathologies, every corridor in D crosses B exactly once. This, together with the fact that $a(R) > 0$, implies that there exists a square s of R that is adjacent to some square of B .

If s is adjacent to two non-adjacent squares t and t' of the arc, then D contains a pathology, which is a contradiction. Indeed, suppose e is the edge shared by s and t and e' is the edge shared by s and t' . If e and e' are opposite edges of s , there

is a double crossing between B and the corridor perpendicular to B at t . If e and e' are adjacent edges in s , then B self-osculates.

Therefore, there are only two cases:

- There is a square of R which is adjacent to two adjacent squares of B , or
- every square of R adjacent to B shares at most one edge with one of the squares of B .

Since self-osculation is forbidden, the first case yields a tessellated hexagon that admits a hexagonal move that pushes B against R , reducing by 1 the area of R . Formally, D is retessellated to $D' \cup s'$, with s' one of the squares in the new tessellation of the hexagon to which we have performed the hexagonal move. Notice that (1) is satisfied. Indeed, the edges of the hexagon induce three corridors before and after the retessellation. In D , the two corridors that intersect the hexagon which are not B intersect B . In D' , the two corridors induced by the same edges also intersect B' , so B' is outermost with respect to the component of the boundary which does not intersect s' . It is clear that the side of B' that intersects s' is contained in $\partial D'$, so B' is an outermost boundary arc in D' . Now, as s' and R' intersect different sides of B' , it is clear that (2) and (3) hold in the new diagram. Since the boundary of the hexagon remains invariant, property (4) holds and there is still an arc B' induced by the same edges as B (satisfying property (5)). Notice that, due to Lemma 3.31, the new diagram still has no pathologies, so property (6) is satisfied. To conclude, we only need to rule out the second case.

For the second case, we assume that every square of R adjacent to B shares exactly one edge with one of the squares of B . Let s_1 be a square adjacent to B . Let t_1 be the square in B adjacent to s_1 and let C_1 be the corridor perpendicular to B at t_1 . Let C_2 be the corridor perpendicular to C_1 at s_1 and let t_2 be the square in B and C_2 .

Now, for $i \geq 2$, let s_i be the square in C_i adjacent to t_i , let C_{i+1} be the corridor perpendicular to C_i at s_i and let t_{i+1} be the square in B and C_{i+1} .

We claim that for $i \geq 2$, $t_{i+1} \neq t_i$ and t_{i+1} lies in B between t_{i-1} and t_i . Indeed, if $t_{i+1} = t_i$ then the corridor C_{i+1} goes through t_i and s_i , which are adjacent, and the corridor self-osculates, contradicting that there are no pathologies. As C_{i+1} does not intersect C_i twice and they already intersect at s_i , the square t_{i+1} in B must be between t_i and t_{i-1} . Therefore, if t_i and t_{i+1} were not adjacent in B , then t_{i+1} and t_{i+2} get closer in B than t_i and t_{i+1} . For this reason, since B is a corridor of finite length, there is some N such that t_N and t_{N+1} are adjacent. Given that, by the assumption in this case, we have that $s_N \neq s_{N+1}$ and the corridor C_{N+1} self-osculates (as it goes through s_N and s_{N+1} and these squares share a vertex). This gives a contradiction. $\square$

We use the previous result to show that pathologies can be eliminated from squared diagrams. The key ideas are to note that, in this setting, any type of (corridor) pathology contains a double intersection, that the internal area of such double intersection can be reduced to zero, and that this double intersection can be eliminated using hexagonal moves.

Lemma 3.33. *Let D be a squared disc diagram. If D contains a pathology P , then $\bar{P}$, the diagram enclosed by P , contains a double intersection.*

Proof. Suppose P is not a double intersection. Take s a square of P with two opposite sides traversed by the fiber of the pathology. Consider the corridor perpendicular to the corridor of the pathology P at s . Since it crosses the fiber of P once, it must cross it a second time, which means it will intersect the corridor of the pathology at a second square. Thus, diagram $\bar{P}$ enclosed by P contains a double intersection. $\square$

Lemma 3.34. *Let D be a disc diagram over a prismatic presentation. If D contains a double intersection P with internal area equal to zero, then D admits a (possibly empty) sequence of hexagonal moves leading to a collapse which reduces the area of D .*

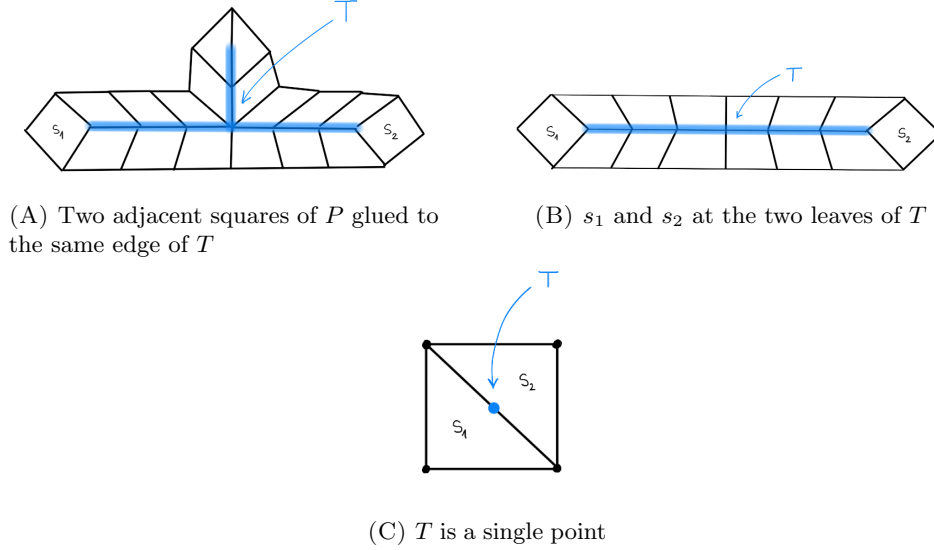
Proof. Let C_1 and C_2 be the two corridors which intersect twice (if they intersect more times, we restrict to two consecutive crossings). Let s_1 and s_2 be the squares at which C_1 and C_2 intersect each other. Because P has internal area equal to zero, the fiber of P contains no 2-cells in its interior region in the sense of Definition 3.25. Let us denote by T the part of ∂P that is contained in the interior region of the fiber of P . Notice that T is a subcomplex of the 1-skeleton of D , so it is a graph. Moreover, T is connected because it can be seen as a retract of P , which is connected. Given that D is simply connected, each loop in its 1-skeleton must be tessellated by 2-cells. Since T lies inside the interior region of the fiber, it must be a connected graph with no loops, therefore a tree. The squares s_1 and s_2 are attached to T at a single vertex, and the rest of squares of P have exactly one side contained in T . Also, since all the edges of T lie in the interior region of the diagram D , they are all adjacent to exactly two 2-cells (which must be 2-cells of P).

If T is a single point, then P consists of two squares glued along two consecutive edges. Due to the small pieces condition (P2) these squares must be induced by the same relation (up to cyclic conjugation), which means they admit a collapse. Suppose then that T contains at least one edge that is a leaf edge. If the two squares of P glued to this leaf are adjacent in P , then D contains again two squares glued along consecutive edges, so the diagram admits a collapse (Fig. 13(A)). If this does not happen, it is necessarily because T has only two leaves and s_1 and s_2 are glued to the vertices of these leaves. This means that P has the form shown in Fig. 13(B) and it admits a series of hexagonal moves that yield a diagram with the same area and containing two squares glued along consecutive edges, which also implies that D admits a collapse. $\square$

Theorem 3.35. *Let D be a pathological squared disc diagram over a prismatic presentation. Then, D admits a (possibly empty) sequence of hexagonal moves and a (non-empty) set of collapses that yields a diagram D' with $\partial D = \partial D'$, with no pathologies and satisfying that two edges in $\partial D'$ induce a corridor if and only if the corresponding edges in ∂D do so.*

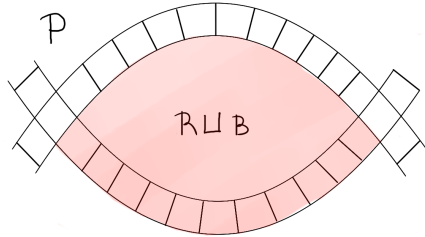
Proof. Let D be a squared disc diagram over a prismatic presentation. If D is pathology-free, we are done. Suppose D contains pathologies. Then we can choose P a pathology in D satisfying that $\bar{P}$, the diagram enclosed by P , contains no pathologies other than P .

By Lemma 3.33, P must be a double intersection of two corridors. Suppose that the internal area of P , say $a_i(P)$, is strictly greater than zero. Then, we may apply Theorem 3.32 taking R to be the internal region of P and B to be the segment

FIG. 13. Different configurations of P in Lemma 3.34

of corridor that remains when we remove from P one of the corridors involved in the double intersection (see Figure 14). Indeed, $R \cup B$ has B as an arc at its boundary and $a(R) = a_i(P) > 0$. Moreover, $R \cup B$ contains no pathologies and B is outermost boundary arc in $R \cup B$ because $\bar{P}$, the diagram enclosed by P , contains no pathologies different from P . Therefore, by this theorem, we conclude that the internal area of P can be reduced without modifying the area of D . This means that we may now assume that the internal area of P is zero. So, by Lemma 3.34, the diagram admits a sequence of hexagonal moves leading to a collapse that yields a diagram D' with smaller area. If the new diagram D' has no pathologies, we are done; if not, we repeat the reasoning recursively. Since in each iteration the area of the diagram decreases 2 units and the initial diagram has finite area, this process will eventually finish with a diagram of area zero or area one (which are always non pathological) or with a diagram of greater area which is non pathological.

The last claim of the theorem follows from Lemma 3.18. $\square$

FIG. 14. Choosing the subcomplexes R and B in the double intersection from Theorem 3.32.

3.4. Existence of prismatic diagrams free of pathologies. Siamese spiders, defined below, are subdiagrams that we may encounter when working with prismatic presentations and that present an obstacle to finding diagrams without spider pathologies. The results in this subsection have in their hypothesis disc diagrams without siamese spiders, and we postpone dealing with these subdiagrams until Section 4.

Definition 3.36 (Siamese spiders). If two spiders S_1, S_2 with heads H_1, H_2 share a leg L , we say that S_1 and S_2 are siamese through the leg L .

Theorem 3.37. *Let D be a disc diagram over a prismatic presentation without siamese spiders. Then, D cannot contain spiders with circular legs or with intersecting legs.*

Proof. Assume that D exhibits pathologies as in the statement, choose among them a pathology P so that $\bar{P}$ does not contain any other circular legs or intersecting legs. If there are any heads in the interior of P , apply prismatic moves to remove them. Note that prismatic moves can not create siamese spiders.

In the following, we will use several results for retessellating squared diagrams that use hexagonal moves and collapses/uncollapses. In view of Lemma 3.18 these results do not create siamese spiders.

Let us now consider the two possible cases.

Assume that P is given by a pair of intersecting legs L_1, L_2 and a head H . We can retessellate the squared subdiagram given by $\bar{P} - H$ using Theorem 3.35, so that the retessellated subdiagram is free of pathologies and the boundary edges of L_1 and L_2 still define intersecting legs L'_1 and L'_2 . We may now apply Theorem 3.32 on the retessellated subdiagram taking R to be the interior region of P and B to be L'_2 : indeed, L'_2 is an outermost boundary arc, if it was not there would be a double intersection between L'_1 and another corridor. The application of Theorem 3.32 leaves us with a diagram where the pathology given by the two intersecting legs L'_1 and L'_2 has an interior region with area zero.

Arguing as in Lemma 3.34 we take the corresponding tree T . If any leaf edges in T correspond to pairs of squares glued along two edges, due to P2 the two squares must be induced by the same relation and can be collapsed, trimming this way the corresponding leaf edge in the tree T . After a finite number of iterations, the tree must be a line and the leaf edge which is not adjacent to the head H must correspond to the intersection square of the two legs. At this point, a sequence of hexagonal moves can be performed to reach a diagram where the intersection square is glued to the head along two consecutive edges, contradicting P2. See Fig. 15.

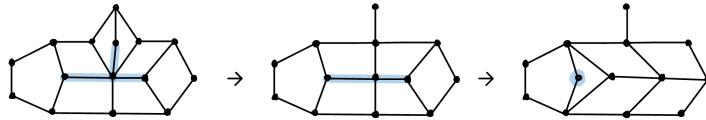


FIG. 15. Trimming tree process for two intersecting legs.

We now know that P could not be given by a pair of intersecting legs, so P is given by a head H and a circular leg L with no heads in its interior. Notice that we may

assume that L is induced by a pair of adjacent edges of the head (otherwise, $\bar{P}$ would contain two intersecting legs or onether circular leg). Let s be one of the squares of L that is adjacent to the head. We can retessellate the squared subdiagram given by $\bar{P} - (s \cup H)$ using Theorem 3.35 in order to eliminate pathologies from this subdiagram while maintaining L . In the retessellated subdiagram, we may apply Theorem 3.32 taking R to be the interior region of the pathology and B to be the retessellated leg L . This gives a diagram where the interior region of the pathology has area zero, we take the corresponding tree T as in Lemma 3.34. Because this pathology only involves one corridor, all of the leaf edges in this tree that are not adjacent to the head H of the pathology must correspond to two squares glued along two consecutive edges. Due to P2, these pairs must come from the same relation and a collapse can be performed. After a finite number of collapses we end up with a tree consisting of an edge one of whose vertices belongs to the head H . Now the circular leg is exhibited by the subdiagram given by the head and the adjacent two squares that are glued along two consecutive edges (Fig. 16). Applying one last collapse to the pathology shows that the head is induced by a relation which was not cyclically reduced, contradicting Definition 2.6. We conclude that a disc diagram over a prismatic presentation cannot exhibit circular legs either.

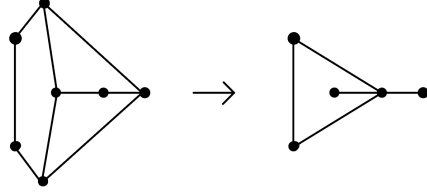


FIG. 16. Collapse of degenerate circular leg gives a not cyclically reduced head.

□

Corollary 3.38. *Let D be a disc diagram over a prismatic presentation without siamese spiders. There is an alternative diagram D' with $\partial D = \partial D'$ such that D' contains no pathologies, spider pathologies or siamese spiders.*

Proof. Using Theorem 3.37, we know that D contains no spider pathologies. If the resulting diagram has no pathologies, we are done. Otherwise, take a pathology P and remove any heads in its interior region using prismatic moves. This shows that the set of retessellations of D with no siamese spiders, spider pathologies and containing a pathology P with no heads in its interior region is non-empty. Among all of these diagrams, choose one with minimal area and call it D' . Let P' be a pathology in D' with no heads in its interior region. Using again Theorem 3.37 we know that P' is a corridor pathology and $\bar{P}'$ is a squared diagram, which means that Theorem 3.35 can be applied to it reducing the area of the pathology and contradicting the minimality in the choice of D' . □

4. SQUASHING SPIDERS AND NON-GEODESICS

Let $\langle \Sigma \mid S \sqcup H \rangle$ be a prismatic presentation. The main applications of this paper (SL-automaticity, FFTP, Tits-rewriting) rely on being able to recognise geodesics in the Cayley graph. In this section, we show that if w is non-geodesic then there is a diagram D that exhibits the non-geodesicity in a strong sense; intuitively, we will see that D has a subdiagram consisting of a corridor or more than half of a spider, and most of the boundary of this subdiagram is in the boundary of D and labels a subword of w . This intuition will be formalised through the concept of an *empty w -arc* (see Fig. 21, Definitions 4.7 and 4.17).

We state now the main technical result of the paper, whose proof is the topic of this section. The missing definitions will be provided afterwards.

Theorem 4.1. *Let $\langle \Sigma \mid S \sqcup H \rangle$ be a saturated prismatic presentation. Let w be a non-geodesic word over $\Sigma \cup \Sigma^{-1}$. Then there exists a disc diagram D over $\langle \Sigma \mid S \sqcup H \rangle$ that contains an empty w -arc.*

4.1. Fusion of heads. The previous section showed that we can prevent the existence of pathologies provided that we start with a diagram without siamese spiders (Corollary 3.38). In order to deal with siamese spiders we introduce *saturated prismatic presentations* $\langle \Sigma \mid S \sqcup H \rangle$, which are prismatic presentations that have an extra property that we call *fusion of heads*.

Definition 4.2 (Saturated prismatic presentation). A prismatic presentation $\langle \Sigma \mid S \sqcup H \rangle$ is *saturated* if it satisfies conditions **P1-P4** together with the following property:

P5 (Fusion of heads) If σ and τ are cyclic permutations of relations in $H \cup H^{-1}$ of the form $\sigma = uv$ and $\tau = v^{-1}w$ with $v \neq \epsilon$, then the cyclic reduction of uw is a cyclic permutation of a relation in $H \cup H^{-1}$ or the empty word.

We remark that a prismatic presentation with cyclic spiderheads is saturated; we now show that graph products of triangular presentations also admit a saturated prismatic presentation:

Proposition 4.3. *Consider $\Gamma = (V\Gamma, E\Gamma)$ a simplicial graph and a collection $\{G_v = \langle G_v - \{1\} \mid R_v \rangle\}$ of triangular presentations associated to the vertices in $V\Gamma$. The graph product $\mathcal{G} = \mathcal{G}(\Gamma, \{G_v\}_{v \in \Gamma})$ of the groups G_v associated to the graph Γ naturally admits a saturated prismatic presentation.*

Proof. The saturated prismatic presentation $\langle \Sigma \mid S \sqcup H \rangle$ associated to the graph product has as its generating set the union of the vertex generating sets,

$$\Sigma = \cup_{v \in V\Gamma} (G_v - \{1\}).$$

The set of squares is also the natural one, given by

$$S = \{ab = ba \text{ for } a \in \Sigma_u, b \in \Sigma_w, \{u, w\} \in E\Gamma\}.$$

In general, taking H to be the set of triangular heads $\cup_{v \in V\Gamma} R_v$ would not satisfy condition **P5**, however, we can complete (saturate) this set until it does. Formally, let $v \in V\Gamma$ and let $H_{v,0} = R_v$. Inductively define $H_{v,i} = H_{v,i-1} \cup F_{v,i-1}$, where the set $F_{v,i-1}$ is given by cyclic reductions of relations of the form $\sigma\tau$ where σ, τ are cyclic permutations of relations in $H_{v,i-1} \cup H_{v,i-1}^{-1}$ with $\sigma = uv, \tau = v^{-1}w$ and $v \neq \epsilon$. By construction, the set of heads

$$H_v = \cup_{i=0}^{\infty} H_{v,i}$$

satisfies **P5** and, because the generating sets associated to different vertices in Γ are disjoint, taking $H = \cup_{v \in V\Gamma} H_v$ ensures that $\langle \Sigma \mid S \sqcup H \rangle$ satisfies **P5**. The fact that this presentation also satisfies conditions **P1-P4** is a consequence of Proposition 2.14, together with the fact that the prisms associated to the fused heads added in H_v to R_v can be seen as the fusion of the prisms associated to the heads in $H_{v,0} = R_v$. $\square$

Note that the saturated prismatic presentation in the previous result does not necessarily have triangular spiderheads.

The saturated presentations that we will be working with, and are worth keeping in mind throughout the text, are of the two types described above.

Remark 4.4. Proposition 2.14 states that the natural presentation of a graph product of prismatic presentations is prismatic. It is also true that the natural presentation of a graph product of saturated prismatic presentations is a saturated prismatic presentation.

The main purpose of the fusion of heads is to remove siamese spiders, which can be done as follows:

Lemma 4.5. *Let D be a disc diagram over a saturated prismatic presentation. If D contains siamese spiders, there is an alternative disc diagram D' with $\partial D = \partial D'$ that contains no siamese spiders.*

Proof. We check that each pair of heads τ_1, τ_2 sharing a leg L can be removed by a retessellation of the diagram. We first perform a (sequence of) prismatic moves along $\tau_1 \cup L$ to “push” τ_1 towards τ_2 . Formally, $\tau_1 \cup L$ gets retessellated to the union of a head τ'_1 and a set of legs, where τ'_1 and τ_2 share an edge. We distinguish two cases now: if τ'_1 and τ_2 are induced by the same relation and the diagram $\tau'_1 \cup \tau_2$ is not reduced (that is, $\partial\tau'_1$ and $\partial\tau_2$ label the same word if read starting at their common edge), we can retessellate $\tau'_1 \cup \tau_2$ to a diagram with area zero given by an edge path. Otherwise, if $\tau'_1 \cup \tau_2$ yield a reduced diagram, Condition **P5** (fusion of heads) ensures that $\tau'_1 \cup \tau_2$ can be retessellated by a single head relation, reducing the number of siamese spiders in the original diagram D . Doing this repeatedly, we obtain the desired diagram D' . $\square$

Corollary 4.6. *Let w be a trivial word in a saturated prismatic presentation. Then there exists a disc diagram with $\partial D = w$ such that D is free of siamese spiders, spider pathologies and corridor pathologies.*

Proof. By Van Kampen’s Lemma, a diagram D for w exists. By Lemma 4.5 we can assume that D has no siamese spiders. Now we finalise using Corollary 3.38. $\square$

4.2. w -arcs. We now introduce w -arcs, subdiagrams that combine the ideas behind arcs and spider arcs (Definitions 3.27 and 3.28) and which we will manipulate in future sections in order to characterise geodesics in saturated prismatic presentations.

Definition 4.7 (w -arc). Let D be a disc diagram over a prismatic presentation containing no spider pathologies or siamese spiders and let w be a subword of the word labelling ∂D . We say that a subdiagram of D is a w -arc if it is either an arc or a spider arc whose extremes lie in w and, in the case of a spider given by an n -gonal head, $\lfloor \frac{n-2}{2} \rfloor$ of its legs end in w between the extremes of the spider arc. In particular, if the w -arc is given by a spider, the majority of its legs end in w .

The *sides* of a w -arc are the sides of the corresponding arc or spider arc.

Notation 4.8. Let D be a disc diagram and let u be a subword of its boundary label. We will write $\partial_u D$ to mean $\partial D \cap u$, where u can be seen as a 1-subcomplex in the boundary of D .

Definition 4.9 (Region enclosed by w -arc). Given a w -arc A , the *region enclosed* by A and $\partial_w D$, denoted I_A , is the region enclosed by $\partial_w D$ and the side of A whose end vertices are closest in $\partial_w D$ along w . Informally, we choose the region I_A so that it does not include A .

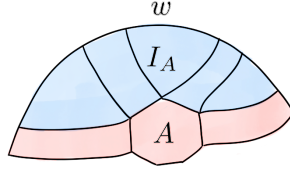


FIG. 17. Spider w -arc A in a disc diagram D , the majority of the spider's legs end in w . The region enclosed by A and $\partial_w D$ is denoted I_A .

Remark 4.10. A w -arc A in a disc diagram D is uniquely determined by the two edges in w where A intersects $\partial_w D$.

Definition 4.11 (Outermost w -arc). Given a diagram D as in Definition 4.7 and a word w along ∂D , we say that a w -arc A is an *outermost w -arc* in D if I_A does not contain other w -arcs.

The following two statements show that the set of w -arcs in a diagram is preserved by both hexagonal and prismatic moves, and their relative containment is preserved in the sense that outermost w -arcs map to outermost w -arcs.

Lemma 4.12 (Prismatic moves preserve w -arcs). *Consider a disc diagram D over a prismatic presentation with no spider pathologies or siamese spiders and a word w along ∂D , and let D' be a disc diagram obtained from D by applying a prismatic move. Two edges f_+, f_- in w determine a w -arc in D if and only if they determine a w -arc in D' .*

Proof. Let $\tau \cup L_1 \cup \dots \cup L_n$ be the half-prism in D to which the prismatic move is applied, where τ denotes a polygon and $L_1, \dots, L_n$ denote the n lateral faces. The prismatic move will retessellate the half-prism to its other half $\tau' \cup L_{n+1} \cup \dots \cup L_m$ where τ' is its other base and $L_{n+1} \cup \dots \cup L_m$ are the remaining lateral faces. For $i = 1, \dots, m$, denote by h_i, h'_i the edges of τ and τ' respectively that are connected by the lateral face L_i in the full prism.

Consider a w -arc A in D with extremes f_- and f_+ in $\partial_w D$. Suppose that A traverses the original half-prism and call e_-, e_+ the two intersection edges in the boundary of the half-prism. If A intersects τ in edges $h_i, h_j \in \tau$ as in Fig. 18(A), the prismatic move will yield a new disc diagram D' where edges f_-, f_+ still determine a w -arc A' which traverses e_- and e_+ and goes through the other base τ' through its edges h'_i and h'_j (Fig. 18(B)). This is a consequence of the fact that the prismatic move can be understood as an elongation or contraction of the legs (lateral faces) of a spider with head τ .

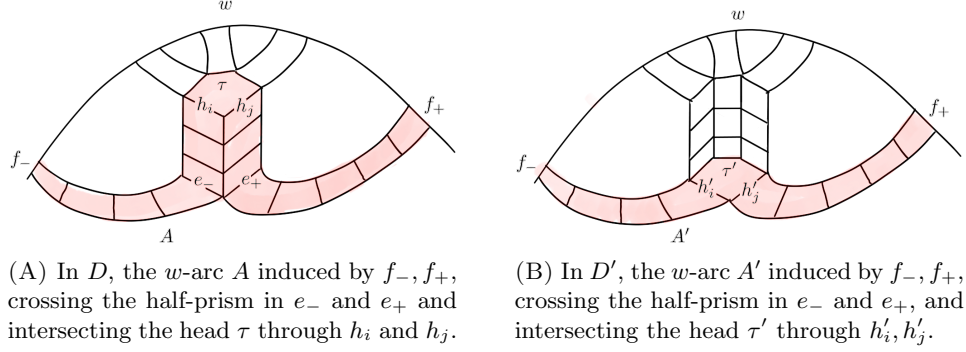


FIG. 18. On the left, a diagram D with a w -arc A traversing the base of a half-prism. A prismatic move yields the diagram D' on the right where A becomes A' and traverses the other base.

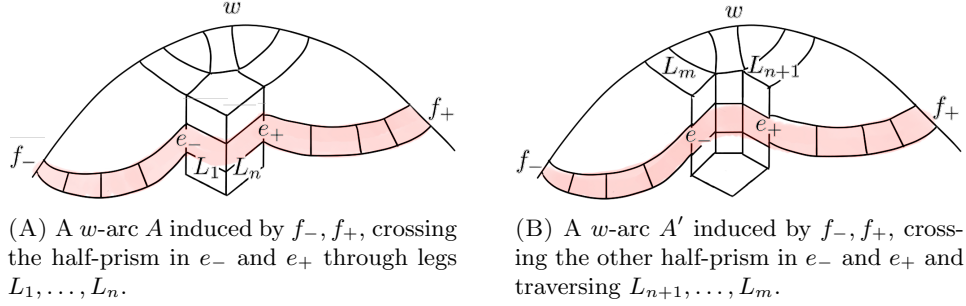


FIG. 19. On the left, a diagram D with a w -arc A traversing the lateral faces of a half-prism. A prismatic move yields the diagram D' on the right where A becomes A' and traverses the other lateral faces.

Keeping the above notation, consider now the case where the w -arc A traverses the half-prism but not τ . This means that A is transversal to the lateral faces $L_1, \dots, L_n$ as in Fig. 19(A), and in this case the prismatic move yields a diagram D' where f_- , f_+ still determine a w -arc A' that traverses e_- , e_+ and is transversal to the remaining lateral faces $L_{n+1}, \dots, L_m$ as Fig. 19(B) shows.

The reasoning above shows that edges f_- , f_+ will only determine a w -arc before [after] the prismatic move if they do so afterwards [before]. Moreover, any w -arcs in D not traversing the retesselated half-prism will not be affected by the prismatic move. $\square$

Lemma 4.13. *Consider a disc diagram D over a prismatic presentation with no spider pathologies or siamese spiders and a word w along ∂D , and let D' be a disc diagram obtained from D by applying a prismatic move. If B and C are w -arcs in D inducing w -arcs B' , C' in D' that satisfy $C \subset I_B$, then it cannot happen that $B' \subset I_{C'}$.*

Proof. Let us see why the containment between w -arcs cannot be inverted by prismatic moves. Suppose B and C are w -arcs in D inducing w -arcs B' , C' in D' that

satisfy $C \subset I_B$. Let f_-, f_+ and g_-, g_+ be the edges in $\partial D = \partial D'$ that determine the arcs B, B' and C, C' respectively. Because $C \subset I_B$, in $\partial_w D$, g_- and g_+ lie between the edges f_- and f_+ . Since prismatic moves do not change the boundary of D and B' and C' are determined by the same edges of $\partial_w D$ as B and C , it cannot happen that $B' \subset I_{C'}$ (for this to happen, we would need to have that f_- and f_+ lie between the edges g_- and g_+ in $\partial_w D$). $\square$

Lemma 4.14 (Hexagonal moves preserve w -arcs). *Consider a disc diagram D over a prismatic presentation with no spider pathologies or siamese spiders and a word w along ∂D , and let D' be a disc diagram obtained from D by applying a hexagonal move. Two edges f_+, f_- in w determine a w -arc in D if and only if they determine a w -arc in D' .*

To see that the previous statement holds, one just needs to notice that the extremes of the three corridors in a hexagon still induce three corridors with the same intersection pattern after a hexagonal move.

Lemma 4.15. *Consider a disc diagram D over a prismatic presentation with no spider pathologies or siamese spiders and a word w along ∂D , and let D' be a disc diagram obtained from D by applying a hexagonal move. Let A and B be w -arcs in D which induce w -arcs A', B' in D' . Then, $B \subset I_A$ if and only if $B' \subset I_{A'}$.*

Proof. That the containment is preserved is trivial if the w -arcs A and B do not intersect the hexagon which is being altered, and also if both A and B intersect the hexagon because that would mean that both A, B and A', B' are intersecting pairs of w -arcs so there is no containment between them. To see that the claim is true if only one w -arc, say A , intersects the hexagon, denote by Δ the hexagon diagram in D and Δ' the hexagon diagram in D' (after the hexagonal move) and notice the following. The subcomplex $A \cup \Delta$ gets retessellated to $A' \cup \Delta'$, and if B doesn't intersect A then B cannot cut through $A \cup \Delta$ either. This means that B' does not cut through $A' \cup \Delta'$, and since D and D' only differ in Δ , we have that $B \subset I_A$ if and only if $B' \subset I_{A'}$. $\square$

LAM: dejamos H, H'
por aquí o cambiamos
también notación? **PLL:**
Pongo Δ en vez de H

The next lemma shows that we can retessellate the region enclosed by an outermost w -arc and the boundary $\partial_w D$ so that this region does not contain any spider heads or spider pathologies and the initial w -arc remains outermost.

Lemma 4.16. *Let D be a disc diagram over a prismatic presentation containing no spider pathologies and no siamese spiders. If D contains an outermost w -arc A , then there exists an alternative disc diagram D' satisfying the following:*

- (1) $\partial D = \partial D'$,
- (2) D' contains an outermost w -arc A' with the same extremes as A ,
- (3) the region $I_{A'}$ enclosed by A' and $\partial_w D$ contains no spider heads (that is, it is a squared diagram),
- (4) if the w -arc A' is given by a spider, its head intersects $\partial_w D$, and
- (5) the region $I_{A'} \cup A'$ contains no corridor or spider pathologies.

Proof. Let $D = D_0$ be as in the statement of the theorem. If I_A contains heads, we may apply prismatic moves in order to take them out: indeed, since D contains no spider pathologies and A is outermost, every head τ in I_A has a leg that intersects A , call it L . This head can be removed from the region enclosed by the w -arc by applying a prismatic move to $\tau \cup L$ in $A \cup I_A$ and, doing this iteratively for every

head inside I_A , we obtain a new diagram D_1 satisfying (1), (2) and (3). Notice that (2) is satisfied by Lemmas 4.12 and 4.13.

Let A_1 be the outermost w -arc in D_1 having the same extremes as A . If A_1 is given by a spider with head τ_1 , we can always perform a prismatic move along one of the legs of the spider arc so as to push the head towards $\partial_w D$. In this way, we obtain a diagram D_2 with outermost w -arc A_2 satisfying (1), (2), (3) and (4). As before, (2) remains true by Lemmas 4.12 and 4.13.

Notice that D_2 contains no spider pathologies by Theorem 3.37. It may happen, however, that the prismatic moves that we have performed have created corridor pathologies in $(I_{A_2} \cup A_2) - \tau_2$, where τ_2 is the head of A_2 (if A_2 is a spider arc). If this is the case, we retessellate $(I_{A_2} \cup A_2) - \tau_2$ using Theorem 3.35 to obtain a new diagram D_3 with no pathologies. Observe that, because this retessellation does not change the corridors induced by pairs of edges in the boundary, D_3 contains an outermost w -arc A_3 induced by the same boundary edges as A_2 . Now, because $I_{A_3} \cup A_3$ is a retessellation of the union of $(I_{A_2} \cup A_2) - \tau_2$ (which does not contain corridor pathologies) and the head τ_2 , and taking into account Theorem 3.37, we see that $D' = D_3$ satisfies all five properties in the statement of the theorem. $\square$

We finally give our key definition.

Definition 4.17. Let w be a word over the standard generators of prismatic presentation and consider a disc diagram D with an outermost w -arc A satisfying (1)-(5) of Lemma 4.16. If A is a corridor w -arc, we say that A is *empty* if the area of I_A is null. If A is a spider w -arc given by a spider $\mathfrak{X}$, we say that A is *empty* if the area of $I_A - \mathfrak{X}$ is null.

4.3. Proof of Theorem 4.1. Our next goal is to see that, given a disc diagram with an outermost w -arc satisfying (1)-(5) of Lemma 4.16 above, we can transform it so as to “erase” the area enclosed above the w -arc. We will first show how this is possible if the w -arc is given by a spider and then if it is a corridor.

We establish the following notation:

Notation 4.18. Let D be a disc diagram over a prismatic presentation and let w be a word along its boundary. Suppose that D has an outermost, spider w -arc A with head τ as described in Lemma 4.16. We introduce the following notation to describe the spider (see Fig. 20).

We denote by w_0 the (possibly trivial) subword of w that labels $\tau \cap \partial_w D$ (it is not difficult to see, from the definition of arc, that $\tau \cap \partial_w D$ is a non-empty, connected subcomplex of $\partial_w D$). If w_0 is given by a single letter we may also refer to it as L_0 referencing the (trivial) leg this edge represents. We may now write $w = w'w_0w''$ and we can further factorise

$$\begin{aligned} w' &= w_{-(p+1)}f_{-p}w_{-p}f_{-(p-1)} \cdots f_{-2}w_{-2}f_{-1}w_{-1}, \text{ and} \\ w'' &= w_1f_1w_2f_2 \cdots w_qf_qw_{q+1}, \end{aligned}$$

where each f_i is an edge where one of the spider’s legs intersects $\partial_w D$ and w_i are the remaining subwords in the partition induced by the edges f_i . We denote by L_i the leg of the spider which has f_i as an extremal edge and, for $i \in \{1, \dots, q\}$, the side of L_i that shares a vertex with w_i will be called the upper side u_i and the side of L_i that shares a vertex with w_{i+1} will be called the lower side l_i . Symmetrically for $i \in \{1, \dots, p\}$, the side of L_{-i} that shares a vertex with w_{-i} will be called u_{-i} and the side of L_{-i} that shares a vertex with $w_{-(i+1)}$ will be called l_{-i} .

LAM: andrew has made comments that we often use “as i Lema 4.16” and it is not clear if its as in the hypoth, as in the conclusion... we name this 5 times so he suggests having a name for the spider satisfying (1)-(5). It is not a squashed spider... only head has been squashed
PLL: No se me ocurre ningún buen nombre. Yo lo dejaría como está aunque sea un poco repetitivo.

The region enclosed by u_1 and w_1 is denoted R_1 and, for $i \in \{2, \dots, q\}$, the region enclosed by u_i , w_i and l_{i-1} is denoted R_i . Symmetrically, the region enclosed by u_{-1} and w_{-1} is denoted R_{-1} and, for $i \in \{2, \dots, p\}$, the region enclosed by u_{-i} , w_{-i} and $l_{-(i-1)}$ is denoted R_{-i} . We convene that R_0 is the region enclosed by w_0 and $\partial_w D$, which has null area. The notation is clarified in Fig. 20.

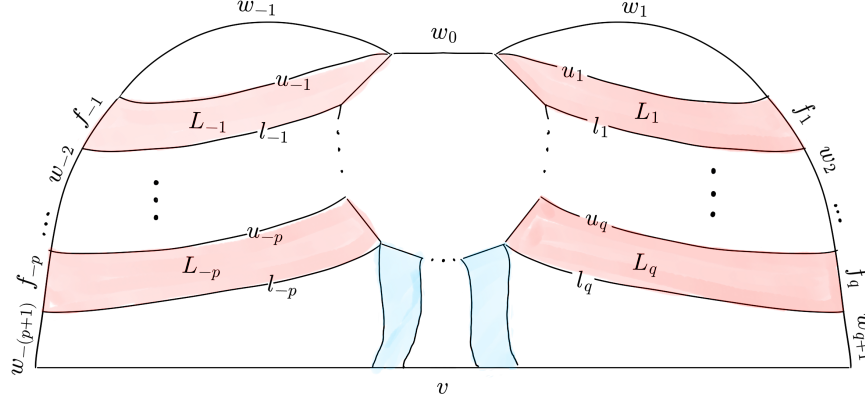


FIG. 20. Spider w -arc in a disc diagram D for a word wv^{-1} where the spider's head intersects $\partial_w D$: each leg L_i in the w -arc as an upper side u_i and a lower side l_i , the end edges of the legs break the path w into a concatenation of f_i -edges and w_i -paths, and the region enclosed by w_i , u_i and l_{i-1} is denoted by R_i .

Definition 4.19. Given $n, m \in \mathbb{Z}$ with $m < n$, a word v over the standard generators of a saturated prismatic presentation representing the trivial element and a subword w of v , we define the set $\mathcal{D}_{m,n}(w, v)$ as the set of disc diagrams for v with an outermost spider w -arc A as in Lemma 4.16 whose regions R_i have null area for $m \leq i \leq n$ (Notation 4.18). For $n \geq 0$ we use the notation $\mathcal{D}_n$ as a shorthand for $\mathcal{D}_{0,n}$ and for $m < 0$ we use $\mathcal{D}_m$ as a shorthand for $\mathcal{D}_{m,0}$.

Proposition 4.20. Let v be a word over the standard generators of a saturated prismatic presentation representing the trivial element and let w be a subword v . Suppose that $\mathcal{D}_0(w, v)$ is non-empty, that is, there is a diagram D with an outermost spider w -arc A as in Lemma 4.16, and that this spider has legs $L_{-p}, \dots, L_{-1}, L_1, \dots, L_q$ intersecting $\partial_w D$ as in Fig. 20. The w -arc A (and therefore the diagram D) can be sequentially altered to obtain a diagram D'' with a w -arc A'' satisfying the requirements for D'' to belong to $\mathcal{D}_{-p,q}(w, v)$. In particular, if $\mathcal{D}_0(w, v)$ is non-empty then $\mathcal{D}_{-p,q}(w, v)$ is non-empty.

Proof. We argue inductively. If the diagram D does not belong to $\mathcal{D}_q(w, v)$, let n be the smallest positive integer such that the region R_n of the w -arc A has non-zero area.

We will see that D admits a hexagonal move that reduces the area of R_n . The subcomplex $D_n \subset D$ given by $R_n \cup L_n$ is a squared diagram that has the leg L_n as an outermost boundary arc, contains no pathologies and none of its corridors begins and ends in $\partial D_n \cap R_n$ – this is due to A being outermost and $A \cup I_A$

exhibiting no pathologies. Theorem 3.32 allows us to push a square of R_n through L_n , retessellating D_n to $D'_n \cup s$ where:

- (1) The subcomplex D'_n satisfies $D'_n = L'_n \cup R'_n$ with L'_n being an arc at its boundary,
- (2) The subcomplex L'_n is a spider leg induced by the same extremal edges as the leg L_n ,
- (3) The area $a(R'_n)$ is less than $a(R_n)$, and
- (4) The retessellation D_n to $D'_n \cup s$ induces a retessellation D to D' where the outermost w -arc A gets transformed to an outermost w -arc A' with the same end-edges in the boundary (Lemmas 4.14 and 4.15), and the square s belongs now to the region R'_{n+1} of A' .

Note that, since this retessellation only affects the leg L_n and this leg is preserved, the new arc A' is given by a spider with the same leg structure as A , say $L'_{-p}, \dots, L'_{-1}, L'_1, \dots, L'_q$. Furthermore the retessellation introduces no pathologies or w -arcs, so A' satisfies the conditions for D' to belong to $\mathcal{D}_n(w, v)$.

By repeated application of Theorem 3.32 we can build a disc diagram starting from D that contains an outermost w -arc which, having the same leg structure as A , ensures that the disc diagram belongs to $\mathcal{D}_q(w, v)$. We can now proceed symmetrically on the remaining (“negative”) legs to transform this diagram into a new one D'' with an outermost w -arc A'' that, preserving the leg structure of A , satisfies the conditions for D'' to belong to $\mathcal{D}_{-p,q}(w, v)$. $\square$

The above result shows how to progressively empty a spider w -arc, we do the same with a corridor w -arc.

Proposition 4.21. *Let D be a disc diagram over a prismatic presentation and let w be a subword of ∂D . Suppose that D has an outermost w -arc A which is a corridor and whose interior region I_A is squared and free of pathologies. If D minimises the area of I_A , then A must be an empty w -arc.*

Proof. If $a(I_A) > 0$, we may apply Theorem 3.32 taking $R = I_A$ and $B = A$. Indeed, given that the w -arc A is outermost, there are no corridors beginning and ending at $\partial(I_A \sqcup A) \cap I_A$, so all the hypotheses of the theorem are satisfied. Thus, $I_A \sqcup A$ admits a hexagonal move that yields a retessellation $D' \sqcup s$ of $I_A \sqcup A$ where there is a new outermost w -arc A' satisfying that $I_{A'}$ is squared and free of pathologies and satisfies that $a(I_{A'}) < a(I_A)$, contradicting the minimality assumption. $\square$

Now, we conclude:

Theorem 4.22. *Let D be a disc diagram over a saturated prismatic presentation and let w be a subword of ∂D . If D contains an outermost w -arc, there is an alternative disc diagram D' with $\partial D = \partial D'$ containing an empty w -arc. Moreover, if D is free of siamese spiders, the empty w -arc in D' is induced by the same end-edges as the original one.*

Proof. If D contains siamese spiders, these can be removed using Lemma 4.5. The original w -arc might not be outermost after the retessellation, but in this case a new outermost w -arc could be found in its interior region. Now, the statement is a direct consequence of Lemma 4.16, Propositions 4.20 and 4.21. $\square$

Finally, we can give a proof of Theorem 4.1.

Proof of Theorem 4.1. Let w be non geodesic, so there is a word u with $u =_G w$ and $|u| < |w|$. By Corollary 4.6, wu^{-1} admits a disc diagram free of pathologies and siamese spiders over the saturated prismatic presentation. The result will follow from Theorem 4.22 once we prove that there exists an outermost w -arc in D .

Suppose now that $w = x_1x_2 \cdots x_n$ and $u = y_1y_2 \cdots y_m$ with $n > m$ and x_i, y_j are elements of $\Sigma \cup \Sigma^{-1}$. Let $C_1, C_2, \dots, C_k$ be the corridors in D which intersect ∂D twice, together with the spiders in D (all of whose legs end in ∂D because there are no siamese spiders). Each edge in the boundary of D is an extreme of some C_i . For every $i \in \{1, \dots, k\}$, if we denote by n_i the number of edges in $\partial_w D$ which are extremes of C_i and by m_i the number of edges in $\partial_u D$ which are extremes of C_i , the following equalities must hold:

$$\begin{aligned} n &= n_1 + n_2 + \cdots + n_k, \\ m &= m_1 + m_2 + \cdots + m_k. \end{aligned}$$

If $n_i \leq m_i$ for all $i \in \{1, \dots, k\}$, then $n = n_1 + n_2 + \cdots + n_k \leq m_1 + m_2 + \cdots + m_k = m$, which is a contradiction. So there exists $j \in \{1, \dots, k\}$ such that $n_j > m_j$. If C_j is a spider, the conclusion is obvious. If C_j is a corridor, it also holds that $n_j + m_j = 2$, so it must be that $n_j = 2$ and $m_j = 0$, and C_j crosses $\partial_w D$ twice. Therefore, D contains a w -arc. $\square$

Theorem 4.23 (Geodesic characterisation). *Let w be a word over the generators of a saturated prismatic presentation. Then, w is geodesic if and only if no disc diagram that is free of pathologies and siamese spiders contains a w -arc.*

Proof. Assume w is geodesic and, for a contradiction, assume that there is a word u with $u =_G w$ for which wu^{-1} admits a disc diagram free of pathologies and siamese spiders that contains a w -arc. By Theorem 4.22, this w -arc A can be assumed to be empty. It is easy to verify that the subpath w_A lying in $A \cap \partial_w D$ is longer than the complement of w_A in ∂A (a finer analysis with proofs will be done in Propositions 5.7 and 5.14). This can be used to produce a shortcut giving then a path shorter than w with the same endpoints.

On the other hand, if w is not geodesic, use Theorem 4.1 to find a diagram with a w -arc. $\square$

5. APPLICATIONS

5.1. Injectivity of the subgroup defined by spiderhead presentations. The first application of spiderhead presentations will be the following:

Theorem 5.1. *Let $G = \langle \Sigma \mid S \sqcup H \rangle$ be the saturated prismatic presentation corresponding to one of the following (i) a prismatic presentation with cyclic spiderheads, or (ii) a graph product of triangular presentations as in Proposition 4.3, then for any head τ the natural map $\langle \Sigma_{H(\tau)} \mid H(\tau) \rangle \rightarrow G$ is injective.*

Proof. Let w be a word on $\Sigma_{H(\tau)}$ such that $w =_G 1$. Let D be a disc diagram for w with no corridor pathologies, spider pathologies or siamese spiders (Corollary 3.38). We will show that D is tessellated only by relations in $H(\tau)$. By the choice of D , every head in D has a leg ending at ∂D . For each head τ , perform a prismatic move along said leg so that we obtain an alternative tessellation in which each corresponding head τ' is at the boundary of D' . Note that if D contains any squares, D' does as well.

LAM: he cambiado la referencia a Proposition 5.11 por una referencia a Proposition 5.14, ya que la primera habla solo de k -locally excluding. Que paloma confirme que está de acuerdo con el cambio **PLL:** Creo que está correcto.

In the case of cyclic spiderheads, the boundary of D (and D') is a word over the alphabet $\{a, a^{-1}\}$, where $\Sigma_{H(\tau)} = \{a\}$. One can consider the diagram $D' - \tau'$ and observe that its boundary is still a word over $\{a, a^{-1}\}$. Iteratively removing all of the heads in D' , we obtain a disc diagram whose boundary label is still a word that is a power of a . Now, if the diagram contains squares (which is the case only if D' contained any), there must be at least one that has two consecutive edges as part of the boundary. This means that this square could be glued to a cyclic head along two edges, contradicting Item **P2**.

In the case of graph products of triangular presentations, the boundary of D (and D') is labelled by the generators $\Sigma_v \cup \Sigma_v^{-1}$ associated to a vertex v in the graph. As heads here are also labelled by $\Sigma_v \cup \Sigma_v^{-1}$, the same will be true for $D' - \{\tau'\}$. Iteratively removing all of the heads in D' , we obtain a disc diagram whose boundary label is still labelled by $\Sigma_v \cup \Sigma_v^{-1}$. Now, if the diagram contains squares (which is the case only if D' contained any), there must be at least one that has two consecutive edges as part of the boundary. This is a contradiction because all squares in this presentation are commutations between elements that belong to different vertices of the graph. $\square$

5.2. Falsification by fellow traveler property.

Definition 5.2 (k -locally-excluding group). Let G be a group, Σ a generating set of G and $k > 0$. The pair (G, Σ) is *k -locally excluding* if a word $w \in (\Sigma \cup \Sigma^{-1})^*$ is geodesic if and only if every subword of length k is geodesic.

We remark that for any cyclic group $G = \langle a \rangle$, $(G, \{a\})$ is k -locally excluding with $k = 2$ if G is infinite or otherwise k is greater than half of the order of G .

Also for any group G , $(G, G - \{1\})$ is 2-locally excluding. We also make the following definition:

Definition 5.3 (k -locally-excluding spiderheads). For a head $\tau \in H$, the spiderhead $H(\tau)$ of $\langle \Sigma \mid S \sqcup H \rangle$ is *k -locally excluding* if $(\langle \Sigma_{H(\tau)} \rangle, \Sigma_{H(\tau)})$ is k -locally excluding.

PLL: reworted

The examples given before show that cyclic spiderheads and triangular spiderheads are k -locally excluding for some k . Moreover, spiderheads in the saturated prismatic presentation given in Proposition 4.3 for graph products of triangular presentations are also k -locally excluding for some k . A theorem [14] of Gilman, Hermiller, Holt and Rees shows that a group G is virtually free if and only if it has a finite generating set Σ such that (G, Σ) is k -locally excluding.

In this subsection we show the following:

Theorem 5.4. *Let $k > 0$ and $G = \langle \Sigma \mid S \sqcup H \rangle$ be a saturated prismatic presentation with k -locally excluding spiderhead presentations. Then, any non-geodesic word $w \in (\Sigma \cup \Sigma^{-1})^*$ asynchronously k -fellow travels with a shorter word representing the same element.*

The property of the conclusion of the theorem is known as the *falsification by fellow traveler property* (FFTP for short). This is a property introduced by Neumann and Shapiro [26] that implies, when the generating set is finite, that the language of geodesics is regular and that the growth is rational. See [3] for further discussions. We recall the definition of fellow travelling for completeness.

Definition 5.5 (Fellow traveller for graphs). For $K \geq 0$, we say that two paths $p = v_1 \dots v_n$, $q = u_1 \dots u_m$ in a graph X *asynchronously K -fellow travel* if there are non-decreasing maps $f: [0, n] \rightarrow [0, m]$, $g: [0, m] \rightarrow [0, n]$ satisfying that the vertices $v_i, u_{f(i)}$, as well as the vertices $u_i, v_{g(i)}$ are at distance at most K in the graph for all $i \in [0, n]$ and $i \in [0, m]$ respectively. We say that p and q *synchronously K -fellow travel* if v_i, u_i are at distance at most K for all $i \geq 0$. Notice that for $i \geq n$ and $j \geq m$, $v_i = v_n$ and $u_j = u_m$.

We are ready to prove Theorem 5.4. Essentially, we need to consider the two types of empty w -arcs that exist for a non-geodesic (Theorem 4.1) and use them to find a fellow-traveling shortcut.

Definition 5.6 (Shortcut for empty corridor w -arc). Let A be an empty corridor w -arc in a disc diagram D over a prismatic presentation. Let e, e' be the edges in the equivalence class of the corridor that are contained in $\partial_w D$, let u_A be the side of the corridor running along $\partial_w D$ and let l_A be the other side of the corridor A . For the subword of w given by

$$w_A = e \cdot u_A \cdot e'$$

we define the *shortcut for w_A through A* , denoted α_A , to be the path with the same endpoints as w_A given by l_A .

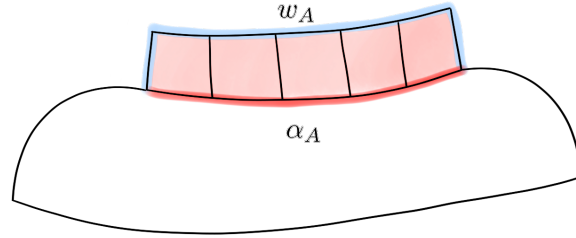


FIG. 21. Shortcut for an empty corridor w -arc A .

The next follows directly from definitions

Proposition 5.7. *For an empty corridor w -arc A , the shortcut α_A for w_A is strictly shorter than w_A and 1-fellow travels with it.*

Lemma 5.8. *Let $D \cup B$ be a disc diagram over a saturated prismatic presentation $\langle \Sigma \mid S \sqcup H \rangle$, where D is a diagram for vu^{-1} , B is a corridor arc at the boundary of $D \cup B$, $\partial D \cap \partial B = v$ and u is geodesic. This diagram can be extended to a prism that has the polygon defined by D as one of its bases and the band B as a subset of its lateral faces.*

Proof. Firstly, because the presentation is saturated and we may assume that D contains no siamese spiders, and because u is geodesic, every head in D has at least one leg ending in B . Using prismatic moves if necessary, we may assume that all heads in the diagram are adjacent to B and apply a prismatic move to the prism induced by each head together with an adjacent lateral face. Let T be the union of all the heads in D and T' the union of all the opposite bases, let L be the set of lateral faces of T intersecting B and L' the remaining lateral faces. Let D_1 be the squared diagram given by $D - T$ and set $B_1 := (B - L) \cup L'$. Notice that $\partial D_1 \cap u$

is a subword of u and therefore geodesic and that we may assume that D_1 is free of pathologies using Corollary 3.38.

We will now see that a hexagonal move can always be performed involving two squares in B_1 and a square s_1 in D_1 . If $D_1 \cup B_1$ is free of pathologies, Theorem 3.32 ensures that there is a square s_1 of D_1 that shares two consecutive edges with B_1 . If $D_1 \cup B_1$ contains pathologies, these must be double crossings involving B_1 . Call w the side of B_1 adjacent to D_1 . There must be a double crossing where one of the corridors defines an outermost w -arc in D_1 . This w -arc can therefore be emptied using Theorem 3.32 without creating pathologies, and give a square of D_1 , call it s_1 , that shares two consecutive edges with B_1 .

Now, the lateral faces L_1 and L_2 in B_1 corresponding to these two adjacent edges together with s_1 induce a tessellated hexagon that can be completed to a cube. After performing the corresponding hexagonal move, we set $B_2 := B_1 - (L_1 \cup L_2) \cup (L_3 \cup L_4)$, where L_3, L_4 are the opposite faces to L_1, L_2 in the cube. Now let $D_2 := D_1 - s$ and repeat the process (which will terminate because $\text{area}(D_{i+1}) < \text{area}(D_i)$, for every $i \geq 1$).

The prism that extends the initial diagram will be as follows:

- One base will be D .
- The other base will be the union of T' with the set of squares opposite to the set $\{s_i\}_{i \geq 1}$ in the applications of hexagonal moves.
- The lateral faces will be the initial ones in B together with the band given by the set of squares that intersect $\partial_u D$ and belong to B_i for some $i \geq 1$.

□

Lemma 5.9. *Let $\langle \Sigma \mid S \sqcup H \rangle$ be a saturated prismatic presentation. Let $\mathfrak{X}$ be a spider in a disc diagram with head label $\tau \in H$ and let L be a leg of $\mathfrak{X}$. Let v be a subword of τ . Apply a prismatic move along L to get a new spider $\mathfrak{X}'$ with head label τ' and let v' be the unique subword of τ' inducing the same corridor legs as v . Then v is geodesic in $(\langle \Sigma_{H(\tau)} \rangle, \Sigma_{H(\tau)})$ if and only if v' is geodesic in $(\langle \Sigma_{H(\tau')} \rangle, \Sigma_{H(\tau')})$.*

Proof. We will see that if v is non-geodesic, v' is non-geodesic. By symmetry, this will show that if v' is non-geodesic v is non-geodesic and the proof will be complete.

Let u be a geodesic word over $\Sigma_{H(\tau)} \cup \Sigma_{H(\tau)}^{-1}$ with $u = \langle \Sigma_{H(\tau)} \rangle v$, and consider a disc diagram D over $\langle \Sigma \mid S \sqcup H \rangle$ for vu^{-1} .

Since $\langle \Sigma \mid S \sqcup H \rangle$ satisfies P4, consider the prism induced by τ and L whose other base is τ' , and let $L_1, \dots, L_{|v|}$ be the lateral faces adjacent to v . Taking $B = L_1 \cup \dots \cup L_{|v|}$, we can apply Lemma 5.8 to $D \cup B$ to extend this prism to a prism with bases D and D' that, together with the prism with bases τ and τ' , will give a prism that has bases the polygon defined by $\partial(D \cup \tau)$ and the corresponding one defined by $\partial(D' \cup \tau')$.

Let u' be the subword of $\partial(D' \cup \tau')$ whose edges induce the same lateral faces as the edges in u . Since $|u'| = |u| < |v| = |v'|$, we have that v is non-geodesic. □

Definition 5.10 (Inner shortcut for empty spider w -arc). Let A be an empty spider w -arc in a disc diagram D over a saturated prismatic presentation with k -locally excluding spiderheads.

Let τ be the label of the head of A with legs $L_{-p}, \dots, L_q$ intersecting $\delta_w D$ as in Notation 4.18. Let v be the subword of τ consisting of the edges of the head that induce (possibly empty) corridors ending in $\partial_w D$. Since A is a w -arc, we

have that $|v| > \frac{1}{2}|\tau|$. In particular, v is a word in $\Sigma_{H(\tau)}$ and it is non-geodesic in $(\langle \Sigma_{H(\tau)} \rangle, \Sigma_{H(\tau)})$, so there exists a subword v' of v of length at most k that is not geodesic. By applying prismatic moves to push the head of A to $\partial_w D$, and in view of Lemma 5.9, we can further assume that there is a subpath γ of the boundary of the head that has length at most k , intersects $\partial_w D$ and is non-geodesic. Choose a word u in $(\Sigma_{H(\tau)} \cup \Sigma_{H(\tau)}^{-1})^*$ that is shorter than the label of γ and represents the same element in $\langle \Sigma_{H(\tau)} \rangle$.

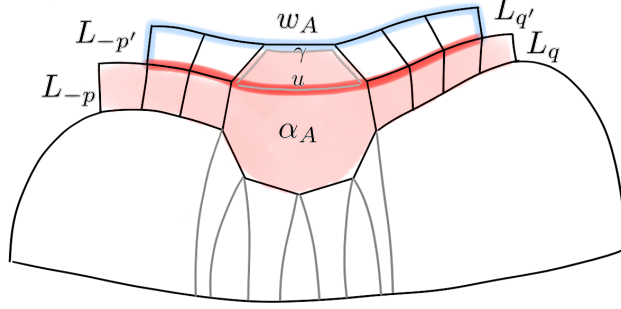


FIG. 22. Inner shortcut for an empty spider w -arc A determined by the legs L_{-p} and L_q . The portion of the head γ lying between $L_{-p'}$ and $L_{q'}$ has a non-geodesic label, that can be shortened to u .

Let $L_{-p'}$ and $L_{q'}$ be the first and last legs of A meeting γ , and let w_A be the subword w_A of w given by

$$w_A = f_{-p'} w_{-p'} \dots w_0 \dots w_{q'} f_{q'}.$$

We define the *inner shortcut* for w_A through A , denoted α_A , to be the path with same endpoints as w_A given by the concatenation $l_{-p'} \cdot u \cdot l_{q'}$ where $l_{-p'}$ and $l_{q'}$ are the lower sides of the legs $L_{-p'}$ and $L_{q'}$ respectively.

The following statement and its proof follow Notation 4.18, in particular for the definition of the subwords w_i , l_i :

Proposition 5.11. *Let D be a disc diagram over a prismatic presentation with k -locally excluding heads. Let A be an empty spider w -arc A and let w_A and α_A as in Definition 5.10 be determined by legs $L_{-p'}$, $L_{q'}$. There are bijections between:*

- The set of edges in $\cup_{-p' \leq i < 0} \{w_i\}$ and the edges in $l_{-p'}$,
- The set of edges in $\cup_{0 < i \leq q'} \{w_i\}$ and the edges in $l_{q'}$,
- The set of edges in $\cup_{i < 0} \{w_i\}$ and the edges in l_{-p} , and
- The set of edges in $\cup_{i > 0} \{w_i\}$ and the edges in l_q .

Proof. We prove the bijection for the first two cases, the remaining two follow similarly.

We define a map

$$\pi: \bigcup_{i \in \{-p', \dots, q'\} \setminus \{0\}} \{w_i\} \rightarrow l_{-p'} \cup l_{q'}$$

that assigns, to each edge e along the domain, the corresponding edge $\pi(e)$ along $l_{-p'} \cup l_{q'}$ in its equivalence class.

We first argue that such edge exists. If the equivalence class of e does not intersect an edge in $l_{-p'} \cup l_{q'}$ it must intersect either w_A or the head of the spider w -arc, but none of these options are possible because of A being outermost and the edge e not being one of the f_i -edges.

Moreover, π is bijective. If it was not injective and for two edges e, e' in the domain we had $\pi(e) = \pi(e')$, it would mean that e and e' span the same corridor contradicting the fact that A is outermost. To see that it is surjective, take an edge e along $l_{-p'} \cup l_{q'}$. If the corridor spanned by e does not contain an edge in the domain of the map π it must contain another edge in $l_{-p'} \cup l_{q'}$ or in the head τ of the spider, but this is not possible because it would yield either an intersection between two legs of a spider or a double intersection between a leg and the corridor spanned by e . Notice that, because $A \cup I_A - \tau$ is disconnected, no corridor induced by an edge of $l_{-p'}$ (respectively, $l_{q'}$) can contain an edge in w_i with $i > 0$ (respectively, $i < 0$). As a consequence, the restrictions of π to $\cup_{i < 0} \{w_i\}$ and $\cup_{i > 0} \{w_i\}$ give us the bijections of the statement. $\square$

Following Notation 4.18:

Proposition 5.12. *Let D be a disc diagram over a prismatic presentation with k -locally excluding heads. Let A be an empty spider w -arc A determined by legs $L_{-p'}$, $L_{q'}$, and let w_A and its inner shortcut α_A as in Definition 5.10. Then α_A is shorter than w_A and the two paths asynchronously k -fellow travel.*

Proof. Using the notation in Fig. 22, observe that the subpath γ along the head has length $|w_0| + p' + q' \leq k$ and is strictly longer than the subword u in the shortcut α_A , so it is clear that α_A is shorter than w_A .

Let π be the bijection

$$\pi: \bigcup_{i \in \{-p', \dots, q'\} \setminus \{0\}} \{w_i\} \rightarrow l_{-p'} \cup l_{q'}$$

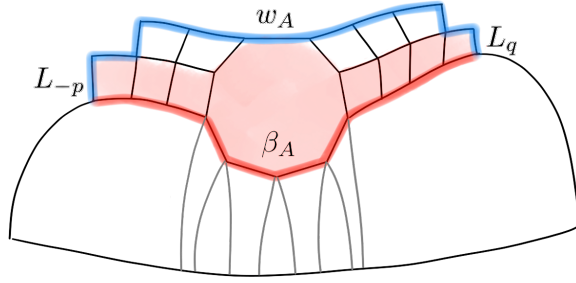
defined in Proposition 5.11. Each vertex in the boundary of an edge in the domain (respectively codomain) of π is at distance at most k – the bound for the length of γ – of its image (respectively preimage). As a consequence, the subword $f_{-p'} w_{-p'} \dots w_{-1}$ asynchronously k -fellow travels with $l_{-p'}$ and a symmetric statement holds for $l_{q'}$. Lastly, each vertex along w_0 is at distance at most $k/2$ of an end-vertex of u , and each vertex along u is at distance at most k of an end-vertex in γ : in conclusion, w_A and α_A asynchronously k -fellow travel. $\square$

The following is an alternative definition of a shortcut for an empty spider w -arc, which has similar properties and which will be used in the settings where we don't have k -locally excluding heads.

Definition 5.13 (Outer shortcut for empty spider w -arc). Let A be an empty spider w -arc in a disc diagram D over a prismatic presentation, where the w -arc is induced by a head τ and legs L_{-p} and L_q as in Notation 4.18. For the subword w_A of w given by

$$w_A = f_{-p} w_{-p} \dots w_0 \dots w_q f_q,$$

we define the *outer shortcut for w_A through A* , denoted β_A , to be the path with same endpoints as w_A given by the concatenation $l_{-p} \cdot h_{-p,q} \cdot l_q$ where l_{-p} and l_q are the lower sides of the legs L_{-p} and L_q respectively and $h_{-p,q}$ is the shortest path along the 1-skeleton of τ between l_{-p} and l_q .


 FIG. 23. Outer shortcut for an empty spider w -arc A .

The outer shortcut has similar properties as the inner one. The main difference is that, without the k -locally excluding heads hypothesis, we cannot ensure a *global* fellow travelling constant as in Proposition 5.11 but instead one that depends on the perimeter of the head involved in the spider arc.

Proposition 5.14. *Let D be a disc diagram over a prismatic presentation. Let A be an empty spider w -arc A and let w_A and its outer shortcut β_A as in Definition 5.13. Then β_A is shorter than w_A and the two paths asynchronously k -fellow travel, where $k \leq \lfloor \frac{n}{2} \rfloor + 1$ and n is the perimeter of the head of A .*

We omit the proof, which follows closely the reasoning of Proposition 5.12 using the last two bijections defined in its Proposition 5.11.

Proof of Theorem 5.4. If w is non-geodesic, take v to be a shorter word over Σ representing the same group element as w . By Theorem 4.23 there is a diagram D for wv^{-1} with a w -arc A , which may be assumed to be empty by Theorem 4.22. Using Propositions 5.7 and 5.12, we see that the shortcut α_A for w_A induces a word u that is shorter than w and k -fellow travels with w as needed. $\square$

5.3. Tits-like rewriting system. We now provide a common generalization for the Tits rewriting system for graph products [15], trickle [4] or t-RAAGs [11].

Definition 5.15 (Tits-like rewriting system). Let $\langle \Sigma \mid H \cup S \rangle$ be a prismatic presentation with triangular spiderheads. We define the following rewriting system for the words over Σ . Letting ε denote the empty word, set $M = M_I \cup M_{II}$, where

$$\begin{aligned} M_I &= \{u \rightarrow v \mid |v| < |u|, uv^{-1} \in H \text{ (up to cyclic permutation/inverse)}\} \cup \\ &\quad \{x^\tau x^{-\tau} \rightarrow \varepsilon \mid x \in \Sigma, \tau \in \{1, -1\}\}, \\ M_{II} &= \{u \rightarrow v \mid |v| = |u|, uv^{-1} \in S \sqcup H \text{ (up to cyclic permutation/inverse)}\}. \end{aligned}$$

For $X \in \{M_I, M_{II}, M\}$, we write $u \xrightarrow{X} v$ if a word v is obtained from a word u in one X -rewriting step, and $u \xrightarrow{X^*} v$ if v is obtained from u in zero or more X -rewriting steps. We say that a word is M -reduced if it cannot be shortened by performing M -rewriting steps.

The rules in M_I reduce the length of a word and are irreversible, whereas the rules in M_{II} maintain the length of a word and are reversible – if $u \xrightarrow{M_{II}} v$, also $v \xrightarrow{M_{II}} u$. We have:

Theorem 5.16. *Let $\langle \Sigma \mid H \cup S \rangle$ be a saturated prismatic presentation with spider-heads and $M = M_I \cup M_{II}$ as above. Then*

- (1) *A word w over Σ is geodesic if and only if it is M -reduced.*
- (2) *Two geodesic words u, v over Σ represent the same group element if and only if $u \xrightarrow{M_{II}^*} v$.*

Proof. (1). If a word w is not M -reduced, it is clearly not geodesic with respect to the generating set Σ .

If w is not geodesic, there is a shorter word v representing the same group element. Take D to be a disc diagram for wv^{-1} over the saturated presentation that is free of pathologies and siamese spiders, such diagram exists by Corollary 3.38. Such a diagram contains a w -arc by Theorem 4.23, and this w -arc can be emptied following Theorem 4.22. A diagram D' for wv^{-1} with an empty w -arc A naturally induces a rewriting $w \xrightarrow{M^*} w'$ that shows that w is not M -reduced. This is because, as defined in Definitions 5.6 and 5.13 if A is given by a corridor (respectively a spider) the shortcut α_A (respectively β_A) can be obtained from the word $w_A \subset \partial_w D'$. More precisely,

$$w_A \xrightarrow{M_{II}^*} w'_A \xrightarrow{M_I} w''_A,$$

where there are as many M_{II} -moves applied as squares in the diagram enclosed by w_A and α_A or β_A in D' .

If A was a corridor w -arc the final M_I -move is of type $x^\tau x^{-\tau} \rightarrow \varepsilon$. Otherwise, if A was w -spider arc, we have that w'_A contains a subword of the label of the head that is not geodesic, and this subword can be substituted by its complement in the boundary of the head by an M_I -move.

This rewriting sequence becomes clear by keeping in mind Figs. 21 and 23.

(2) Assume that u and v are geodesic words over Σ and take D to be a disc diagram over the saturated prismatic presentation for uv^{-1} that is free of pathologies and siamese spiders, which can be done by Corollary 3.38. If the diagram contains no spiders, u and v are only separated by square relations, that is, M_{II} -moves.

If it contains spiders, note that all heads must have even length: an odd head would give rise to a u -arc or a v -arc, contradicting the geodesicity of these words. If D contains an even-legged spider, the number of legs must be equally distributed between $\partial_u D$ and $\partial_v D$ because of u, v being geodesics.

Let x be the last letter in v : because ux^{-1} is not a geodesic, D must contain a ux^{-1} -arc A that we can assume to be empty by Theorem 4.1. Using the ux^{-1} -arc A , we can recursively see that $u \xrightarrow{M_{II}^*} v$ as follows. For the following, we use the notation $w_{[: -1]}$ for the $(|w| - 1)$ -long prefix of a word w .

- (1) If A is a corridor it must have an end in the x -edge, and using its shortcut as in Definition 5.6 we can define an alternative path $u' =_G u$ of the same length as u , having x as its last letter and satisfying $u \xrightarrow{M_{II}^*} u'$ (the M_{II} -moves are precisely the ones provided by the squares in A). The word u' allows us to reduce the problem of understanding if $u \xrightarrow{M_{II}^*} v$ to seeing if $u'_{[: -1]} \xrightarrow{M_{II}^*} v_{[: -1]}$, and we can repeat this process with the diagram D' for $u'_{[: -1]} v_{[: -1]}^{-1}$ that we are left with.
- (2) If A is an empty spider arc it must have an end in the x -edge, and we can use a variation of its outer shortcut (Definition 5.13) to reduce our problem similarly to the corridor case. Using the Notation 4.18, assume that A is given

by a head τ and legs L_{-p}, L_q and let w be the label of the half of τ inducing legs $L_{-p}, \dots, L_{q-1}$ (recall that A is even-legged and half of its legs end in u). Factorising $u = u_0 u_A$, where $u_A = A \cap u$, we have that $u_A \xrightarrow{M_{II}^*} l_{-p} w u_q$, where the M_{II} -moves used come from the squares in the legs $L_{-p}, \dots, L_q$. Moreover, arguing as in the proof of Proposition 5.11, one can see that $|l_{-p} w u_q| = |u_A|$. Now,

$$u_A \xrightarrow{M_{II}^*} l_{-p} w u_q \xrightarrow{M_{II}} l_{-p} w' u_q \xrightarrow{M_{II}^*} u'_A = l_{-p} w'_{[-1]} l_q x,$$

where w' is given by the other half of the head τ and the second (possibly empty) sequence of M_{II} -moves come from the squares in L_q . Taking u' to be $u_0 u'_A$, the problem of understanding whether if u and v are connected by M_{II} moves is again reduced to understanding whether $u'_{[-1]}$ and $v_{[-1]}^{-1}$ are.

Moreover, we have a disc diagram D' for $u'_{[-1]} v_{[-1]}^{-1}$ and $u \xrightarrow{M_{II}^*} u'$.

Note that this process always reduces the perimeter of the diagram involved. By repeated application, we obtain two words u'', v'' and a diagram D'' for $u''(v'')^{-1}$ where $|u''| = |v''|$ and D'' is empty, which implies $u'' = v''$ and trivially $u'' \xrightarrow{M_{II}^*} v''$. By construction, v'' is a prefix of v , so $v = v'' w''$ for some (possibly trivial) word w'' , and $u \xrightarrow{M_{II}^*} u'' w''$ for the same suffix w'' . We conclude that

$$u \xrightarrow{M_{II}^*} u'' w'' \xrightarrow{M_{II}^*} v'' w'' = v.$$

□

5.4. Automaticity of prismatic presentations with cyclic spiderheads. The goal of this section is to prove that groups admitting a finite prismatic presentation with cyclic spiderheads are shortlex automatic, which we define next following [10, Chapters 2.3 and 2.5]. We start recalling the shortlex ordering on words.

Definition 5.17 (Shortlex order). Let $(S, <)$ be a totally ordered alphabet. We will consider two different order relations in S^* :

- The *lexicographical order*: $w_1, w_2 \in S^*$, $w_1 <_{\text{lex}} w_2$ if and only if w_1 is a strict prefix of w_2 or the letter in the first position in which w_1 and w_2 differ is smaller for w_1 .
- The *shortlex order*: for $w_1, w_2 \in S^*$, $w_1 <_{\text{SL}} w_2$ if and only if w_1 is shorter than w_2 or they have the same length and $w_1 <_{\text{lex}} w_2$.

Notation 5.18 ($\text{SL}(g), \text{SL}_G$). The shortlex order is a well-ordering. In particular, if G is a group generated by Σ and we consider the shortlex order on $(\Sigma \cup \Sigma^{-1})^*$ with respect to some ordering of $\Sigma \cup \Sigma^{-1}$, there is a well-defined minimal representative for each element g of the group G under this order, which will be denoted $\text{SL}_{(\Sigma \cup \Sigma^{-1}, <)}(g)$ or simply $\text{SL}(g)$ if $(\Sigma \cup \Sigma^{-1}, <)$ is understood by the context. We will denote by $\text{SL}_{(G, \Sigma \cup \Sigma^{-1}, <)}$ the language of these minimal representatives of elements of G , and write SL_G for short if $(G, \Sigma \cup \Sigma^{-1}, <)$ is understood.

For a definition of a regular language, we refer the reader to [10, Chapter 1].

Definition 5.19. A group $G = \langle S \rangle$ is *shortlex automatic* if SL_G is a regular language and there exists a constant K such that any two words $w, v \in \text{SL}_G$ with $|w^{-1}v|_G \leq 1$ K -fellow travel.

With the previous results and applying [3], we get the following result. Recall that cyclic presentations are k -locally excluding for some k .

Proposition 5.20. *Let G be a finitely generated group with finite generating set Σ . If G admits a finite prismatic presentation with k -locally excluding spiderhead presentations with generating set Σ , then SL_G is regular for any total order on Σ .*

Proof. By Theorem 5.4, G satisfies the falsification by fellow traveller property with respect to Σ . Using the results in [3, Lemma 3.5, Corollary 3.7], this implies that the language of minimal shortlex representatives of elements of G is regular. $\square$

Now, to show that these groups are shortlex automatic, we need to see that there exists a constant K such that any two words $w, v \in SL_G$ with $|w^{-1}v|_G \leq 1$ K -fellow travel. Using the triangle inequality and the fact that w, v are geodesic words, one can see that $|w| \in \{|v| - 1, |v|, |v| + 1\}$: the second case is studied in Proposition 5.22, the first and third case are symmetric and we study them in Proposition 5.23. Without loss of generality, we may assume that w and v have no common prefix – if they had a common prefix z , the words $w = zw'$ and $v = zv'$ would K -fellow travel if and only if the words w' and v' K -fellow travelled.

We fix the following notation for the rest of the subsection.

Notation 5.21. Let G be a group with finite prismatic presentation $\langle \Sigma | H \sqcup S \rangle$ with cyclic spiderheads and fix an ordering on $\Sigma \cup \Sigma^{-1}$.

Proposition 5.22. *With Notation 5.21 there exists a constant K such that for every $w, u \in SL_G$ and $x \in \Sigma \cup \Sigma^{-1}$ with $wx =_G u$ and $|u| = |w|$, w and u K -fellow travel.*

Proof. Let w, u and x be as in the statement and assume that they do not have a common prefix. Let D be a disc diagram for wxu^{-1} over the saturated prismatic presentation of G with no spider or corridor pathologies and no siamese spiders, such diagram exists by Corollary 4.6. We will denote by C_x the corridor induced by x in D .

Suppose that C_x is not a leg of a spider. By symmetry, we may assume that the other extreme of C_x is in $\partial_w D$. Because w is geodesic, Theorem 4.23 implies that there are no w -arcs in the region enclosed by C_x and $\partial_w D$, so C_x is an outermost wx -arc. Thus we may apply Theorem 4.22 to obtain a diagram as shown in Fig. 24, where the word wx can be decomposed as $vew'x$ and $ew'x$ is the boundary of an empty w -arc induced by the edges e and x . The concatenation of v with the shortcut for this empty (wx) -arc gives us a path γ (in blue) with the same endpoints as u and satisfying that $|\gamma| < |w| = |u|$, in contradiction with u being geodesic.

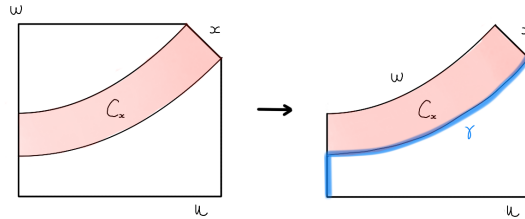


FIG. 24. C_x is not a leg of a spider in Proposition 5.22.

We may then suppose that the corridor C_x in D is a leg of a spider $\mathfrak{X}$. If $\mathfrak{X}$ has an odd number of legs, half of the legs not induced by x end in $\partial_w D$ and the other half in $\partial_u D$ (if not, there would be a w -arc or a u -arc, which gives a contradiction by Theorem 4.23), so $\mathfrak{X}$ gives an outermost wx -arc. If $\mathfrak{X}$ has an even number of legs, half of them end in $\partial_w D$ (or $\partial_u D$) and half minus one of them end in $\partial_u D$ (or $\partial_w D$), for the same reason as before. Without loss of generality, we assume that, if $\mathfrak{X}$ has an even number of legs, half of them end in $\partial_w D$, so $\mathfrak{X}$ gives an outermost wx -arc as well.

Applying Lemma 4.16, we may assume that the head τ of $\mathfrak{X}$ intersects ∂D at the edge x , that the spider still gives an outermost wx -arc (call it A_1), that I_{A_1} contains no spider heads and that $I_{A_1} \cup A_1$ contains no pathologies or spider pathologies. Now, applying Proposition 4.20, we can assume that A_1 is empty in D (see the central diagrams in Fig. 25 and Fig. 26).

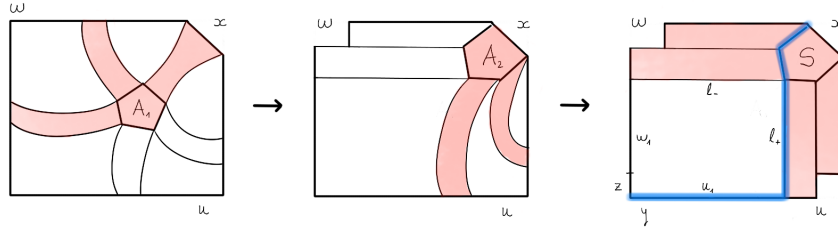


FIG. 25. C_x is a leg of an odd-legged spider in Proposition 5.22.

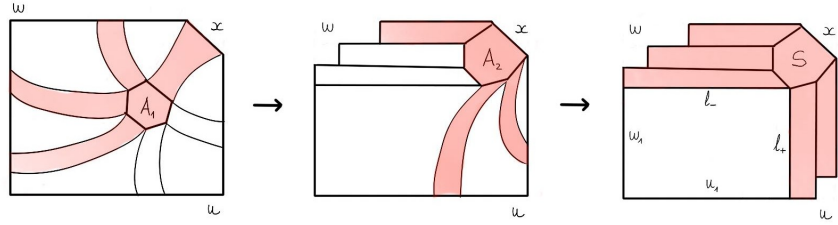


FIG. 26. C_x is a leg of an even-legged spider in Proposition 5.22.

In the odd case, it is clear that the spider induces an outermost xu^{-1} -arc. For the even case, let us denote by w' the suffix of w whose first edge is the end of the leg ending in $\partial_w D$ which is adjacent to the leg induced by x . In this case the spider gives an outermost $w'xu^{-1}$ -arc. Let us call the arc A_2 in both cases. Arguing as we did for A_1 , we may assume that A_2 is empty in D (see the right-most diagrams in Fig. 25 and Fig. 26).

Let us write $w = w_1w_2$ and $u = u_1u_2$, where w_1 and u_1 are the maximal prefixes of w and u , respectively, that are part of $\partial(D - \mathfrak{X})$. Also, let y be the first letter of u and let z be the first letter of w . Moreover, we denote by l_- and l_+ the sides of the legs of $\mathfrak{X}$ that are part of $\partial(D - \mathfrak{X})$ and have a common vertex with w_1 and u_1 , respectively. Applying Proposition 5.11 to the empty arc A_2 , we have that the number of edges in l_+ (in l_-) is equal to the number of edges in u_2 (in w_2) which are not ends of legs of the spider $\mathfrak{X}$.

If $\mathfrak{X}$ has an odd number of legs, by symmetry we may assume that $y < z$. Then, it is clear that reading u_1 , followed by l_+ and, from here, the shortest path in the head $\partial\tau$ with the same endpoint as w , yields a path of the same length as w but lexicographically smaller, in contradiction with $w \in \text{SL}_G$.

If $\mathfrak{X}$ has an even number of legs (say, $2r$). By Theorem 4.23 the region enclosed by u_1l_+ and w_1l_- cannot contain any spiders with an odd number of legs (this would be in contradiction with w and u being geodesics). Therefore, all of the cells tessellating this region have an even number of edges and, since gluing diagrams of even perimeter gives diagrams of even perimeter, this region has even perimeter. However, applying Proposition 5.11, we see that $|u_1l_+| = |u| - (r - 1)$ and $|w_1l_-| = |w| - r$, so the perimeter of the region enclosed by u_1l_+ and w_1l_- is an odd number. This is a contradiction.

Since we reach a contradiction in every possible case, the statement is vacuously true. $\square$

Proposition 5.23. *With Notation 5.21. There exists a constant K such that for every $w, u \in \text{SL}_G$ and $x \in \Sigma \cup \Sigma^{-1}$ with $wx =_G u$ and $|u| = |w| \pm 1$, w and u K -fellow travel.*

Proof. We shall give a proof for $|u| = |wx| = |w| + 1$, and the case $|u| = |w| - 1$ will follow from the symmetry between w and u in these two cases. Without loss of generality, we may assume that w and u have no common prefix.

Suppose then that $|u| = |wx| = |w| + 1$. If the word wx belongs to SL_G , we conclude that $u = wx$ and therefore u and w 1-fellow-travel. If the word $wx \notin \text{SL}_G$, take y to be the first letter of u . This letter is lexicographically smaller than the first letter of w because w and u do not have a common prefix and because $u <_{\text{lex}} wx$.

Let D be a disc diagram for wxu^{-1} over the saturated prismatic presentation of G chosen to be free of corridor and spider pathologies and with no siamese spiders as in Corollary 4.6. We will define a subcomplex A inside D that is an arc or a spider arc running from y to x , we will then show that we can empty the region between A and $\partial_w D$ and the region between A and $\partial_u D$ and, similar to before, this will give us the fellow travelling property that we need. The definition of A is slightly different depending on whether or not the corridor induced by y is the leg of a spider, but the philosophy behind both definitions is the same.

If the corridor induced by y determines a leg of a spider with n legs, notice that n must be even. If n was odd, this spider would induce either a u -arc or a wx -arc but this cannot be the case since both u and wx are geodesic words from Theorem 4.23 ($u \in \text{SL}_G$ and $|u| = |wx|$). Thus, n must be even. Moreover, since D has no u -arcs, at least $\frac{n}{2}$ of these legs end in wx . If no leg ended in x , the spider would induce a $(y^{-1}w)$ -arc B which would be outermost (since w is geodesic, there are no inner w -arcs) that we could empty using Theorem 4.22. Following the notation from Definition 5.13, the shortcut α_B for the word $(y^{-1}w)_B$ would induce a word w' starting with the letter y which would be equivalent to w in the group, with $|w'| \leq |w|$ because $|\alpha_B| < |(y^{-1}w)_B|$ and therefore with $w' <_{\text{lex}} w$. This would contradict the fact that $w \in \text{SL}_G$; thus, this spider has a leg ending in x , a leg ending in y and leaving exactly $\frac{n}{2} - 1$ legs ending in $\partial_w D$ and $\frac{n}{2} - 1$ legs ending in $\partial_u D$. As shown in Fig. 27(B), define the subcomplex A to be the arc given by the head of the spider, the leg induced by y and the leg induced by x .

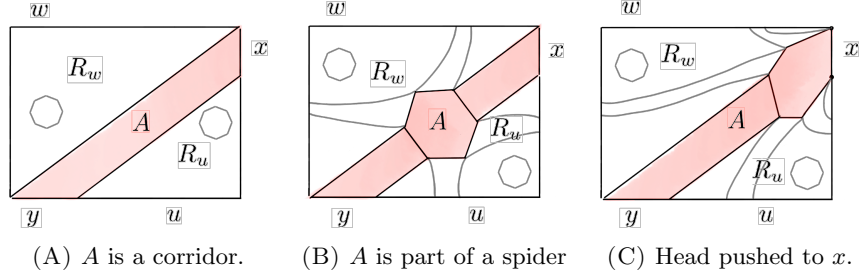


FIG. 27. Definition of the subcomplex A and the regions R_w, R_u (which may contain spider heads) depending on whether the edge y defines a corridor or a spider. If A belongs to a spider, its head can be pushed against the edge x .

If y determines a corridor whose other end is in ∂D , let A be the complex given by this corridor: we will see that A necessarily ends in x . A cannot end in u , because then it would be a w -arc contradicting the fact that u is a geodesic ($u \in \text{SL}_G$). Similar to before, if A ends in w it yields a $(y^{-1}w)$ -arc that can be emptied and from whose shortcut a word w' contradicting the fact that $w \in \text{SL}_G$ can be built. We conclude that the corridor A starting at y must have its other end in the edge x , as in Fig. 27(A).

We now have a subcomplex A , which is both a $(y^{-1}wx)$ -arc and a (yux^{-1}) -arc. Up to a prismatic move we can assume that, if A is given by a spider, its head is pushed against x (see Fig. 27(C)). Now consider the region R_w enclosed between A and $\partial_w D$ and the region R_u enclosed by A and $\partial_u D$, yielding the partition $D = R_w \sqcup A \sqcup R_u$. Since $R_w \cap \partial_w D$ and $R_u \cap \partial_u D$ are labelled by geodesics, the region R_w has no w -arcs and the region R_u has no u -arcs. These regions have no corridor pathologies, spider pathologies or siamese spiders. We can retessellate D to make A into an empty $(y^{-1}wx)$ -arc, and then retessellate the resulting diagram to make the corresponding (yux^{-1}) -arc into an empty one as described in Theorem 4.22.

If A is given by a corridor, the retessellated diagram D (strictly, D' , but we will keep the notation from D for simplicity) will be given by $A \cup R_w$, where R_w is now a union of spider heads that may have been pushed from R_u before emptying the (yux^{-1}) -arc as depicted in Fig. 28(A).

Following Notation 4.18: if A is given by a spider $\mathfrak{X}$ the $(y^{-1}wx)$ -arc involves the legs $L_{-\frac{n}{2}}, \dots, L_0$ and the (yux^{-1}) -arc involves the legs $L_0, \dots, L_{\frac{n}{2}}$ of the spider $\mathfrak{X}$. After emptying both arcs as described above, the retessellated diagram D will be equal to $\mathfrak{X} \sqcup (R_{-\frac{n}{2}-1} \cap R_{\frac{n}{2}+1})$, where $(R_{-\frac{n}{2}-1} \cap R_{\frac{n}{2}+1}) \subset R_w$ is a union of spider heads that may have been pushed from R_u . The resulting diagram is shown in Fig. 28(B).

Regardless of whether or not A is given by a spider, the prismatic moves used in Lemma 4.16 to push heads, together with the absence of siamese spiders in the original diagram, ensure that each of the spider heads left in R_w after emptying the (yux^{-1}) -arc is attached to A through at least one edge: in particular, the heads in R_w cannot pile up and the distance between w and u is uniformly bounded as we detail below.

Let m be the maximal order of a torsion generator in the cyclic spiderhead of $\langle \Sigma \mid S \sqcup H \rangle$ or let $m = 0$ if there are no torsion generators: if A is given by a

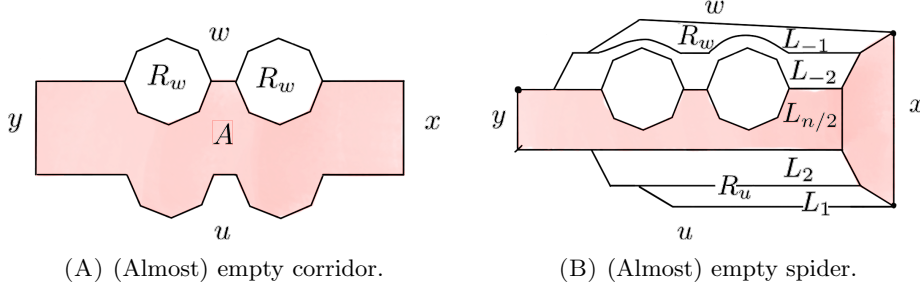


FIG. 28. Once the (yux^{-1}) -arc and the $(y^{-1}wx)$ -arc defined by A are emptied, all the heads from $R_u \cup R_w$ are pushed against R_w and they all intersect A , so w and u K -fellow travel.

corridor then w and u k -fellow travel asynchronously with

$$k = 1 + \left\lfloor \frac{m}{2} \right\rfloor,$$

where the addend 1 is the width of A and the second addend corresponds to crossing the polygons in R_w ; if A is given by a spider of n legs, w and u k -fellow travel asynchronously with

$$k = \frac{n-2}{2} + 1 + \left\lfloor \frac{m}{2} \right\rfloor + \frac{n-2}{2} = (n-1) + \left\lfloor \frac{m}{2} \right\rfloor,$$

where the two addends $\frac{n-2}{2}$ correspond respectively to the distance between u and A and to the distance between w and the polygons in R_w and the addend $1 + \left\lfloor \frac{m}{2} \right\rfloor$ has the same interpretation as in the case where A is a corridor.

Given these observations, it is clear that one can choose a constant K that works for arbitrary choices of words w and u : it is enough to take

$$K = (m-1) + \left\lfloor \frac{m}{2} \right\rfloor, \quad \text{or} \quad K = 1$$

depending on whether the group has torsion generators or not. $\square$

Theorem 5.24. *Every group admitting a finite prismatic presentation with cyclic spiderheads is shortlex automatic.*

Proof. Let G be a group with a finite prismatic presentation $\langle \Sigma | H \sqcup S \rangle$ with cyclic spiderheads. Fix any ordering of the elements of $\Sigma \cup \Sigma^{-1}$. As a consequence of Proposition 5.20, we have that SL_G is a regular language and, by Proposition 5.22 and Proposition 5.23, there exists a constant K such that any two words $w, v \in \text{SL}_G$ with $|w^{-1}v|_G \leq 1$ K -fellow travel. $\square$

It is natural to ask if the SL-automatic structure that we have provided (which is independent of the ordering that we choose for the generators) is also an SL-biautomatic structure. The following example shows that this is not always the case.

Example 5.25. Consider the right-angled Coxeter group given by this presentation

$$G = \langle x, y, z \mid [x, y] = 1, [y, z] = 1, x^2 = y^2 = z^2 = 1 \rangle.$$

If we have that $z < y < x$, the words $u = y^n x z^n$ and $w = z^n y^n$ belong to SL_G and satisfy that $xw =_G u$. Now, for any fellow-travelling constant K , n can be chosen large enough so that u and w do not K -fellow travel.

6. GROUPS WITH A PRISMATIC PRESENTATION: TRICKLE AND T-RAAGs

6.1. Trickle groups. The goal of this section is to show that the standard presentation of trickle groups is a prismatic presentation with cyclic spiderheads, in particular, it is a saturated prismatic presentation: it satisfies conditions **P1-P5** in Definition 2.6.

Trickle groups were informally introduced in Section 2.1, we now give a formal definition following closely [4]. Let Γ be a simplicial graph. The set of vertices of Γ is denoted by $V(\Gamma)$ and its set of edges is denoted by $E(\Gamma)$. The *star* of a vertex $x \in V(\Gamma)$, denoted by $\text{Star}_\Gamma(x)$, is the full subgraph of Γ spanned by $\{y \in V(\Gamma) : \{x, y\} \in E(\Gamma)\} \cup \{x\}$. We will simply write $\text{Star}(x)$ when Γ can be clearly determined from the context.

Definition 6.1. A *trickle graph* is a quadruple $(\Gamma, \leq, \mu, (\varphi_x)_{x \in V(\Gamma)})$, where

- Γ is a simplicial graph,
- $\leq$ is a (partial) order on $V(\Gamma)$,
- μ is a labelling $\mu: V(\Gamma) \rightarrow \mathbb{N}_{\geq 2} \cup \{\infty\}$ of the vertices,
- $\varphi_x: \text{Star}_\Gamma(x) \rightarrow \text{Star}_\Gamma(x)$ is a graph automorphism of $\text{Star}_\Gamma(x)$ for all $x \in V(\Gamma)$.

Let $x, y, z \in V(\Gamma)$. To keep the notation light we write y_x instead of $\varphi_x(y)$; with this convention, $(\varphi_x \circ \varphi_y)(z)$ becomes $(z_y)_x$, which we simplify to z_{yx} . In this way, the expression z_{xyx} represents $(\varphi_{y'} \circ \varphi_x)(z)$, where $y' = \varphi_x(y)$. Moreover, we will write y_{x^a} to refer to $\varphi_x^a(y)$ for $a \in \mathbb{Z}$. As usual, the notation $x < y$ means that $x \leq y$ and $x \neq y$. For $x, y \in V(\Gamma)$, the notation $x \parallel y$ means that x and y are not comparable in the sense that $x \not\leq y$ and $y \not\leq x$. We set $E_{\parallel}(\Gamma) = \{\{x, y\} \in E(\Gamma) : x \parallel y\}$.

The quadruple $(\Gamma, \leq, \mu, (\varphi_x)_{x \in V(\Gamma)})$ must satisfy the following conditions:

- (a) For all $x, y \in V(\Gamma)$, if $x < y$, then $\{x, y\} \in E(\Gamma)$.
- (b) For all $x, y, z \in V(\Gamma)$, if $\{x, y\} \in E_{\parallel}(\Gamma)$ and $z \leq y$, then $\{x, z\} \in E_{\parallel}(\Gamma)$.
- (c) For all $x \in V(\Gamma)$ and all $y, z \in \text{Star}_\Gamma(x)$, we have $z \leq y$ if and only if $z_x \leq y_x$.
- (d) For all $x \in V(\Gamma)$ and all $y \in \text{Star}_\Gamma(x)$, if $y_x \neq y$, then $y < x$.
- (e) For all $x \in V(\Gamma)$, if $\mu(x)$ is finite, then φ_x has finite order and its order divides $\mu(x)$.
- (f) For all $x \in V(\Gamma)$ and all $y \in \text{Star}_\Gamma(x)$, $\mu(y_x) = \mu(y)$.
- (g) For all $x, y, z \in V(\Gamma)$, if $z < y < x$, then $z_{yx} = z_{xyx}$.

We will often say that Γ is a trickle graph meaning that, implicitly, $\leq, \mu$ and $(\varphi_x)_{x \in V(\Gamma)}$ are also given.

Definition 6.2. The *trickle group* $\text{Tr}(\Gamma)$ associated with a trickle graph $(\Gamma, \leq, \mu, (\varphi_x)_{x \in V(\Gamma)})$ is the group defined by the following presentation:

$$\text{Tr}(\Gamma) = \langle V(\Gamma) \mid x^{\mu(x)} = 1 \text{ for } x \in V(\Gamma) \text{ s.t. } \mu(x) \neq \infty, y_x x = x_y y \text{ for } \{x, y\} \in E(\Gamma) \rangle.$$

Remark 6.3. Condition (d) in the definition of a trickle graph ensures that all the relations in the presentation of $\text{Tr}(\Gamma)$ are of the form $xy = zx$ and $x^\mu = 1$. Indeed, for $\{x, y\} \in E(\Gamma)$, we have that the relation $y_x x = x_y y$ becomes

- $yx = x_y y$ if $x < y$,
- $y_x x = xy$ if $y < x$, or
- $yx = xy$ if $x || y$.

Having given the definition of a trickle group, our goal is to prove that conditions **P1-P5** are satisfied by its standard presentation.

- Given a trickle graph Γ , the standard presentation of the trickle group $Tr(\Gamma)$ clearly satisfies condition **P1** (squares and heads) if we let H be the set of relations of the form $x^{\mu(x)} = 1$ and let S be the set of relations associated to quasi-commutations.
- As for condition **P2** (small pieces), observe that an edge in Γ always joins different vertices $x \neq y$ and, therefore, any two adjacent edges in the 2-cell associated to the quasi-commutation relation $x_y y = y_x x$ between x and y must be labelled differently since $\varphi_x(y) = x \Leftrightarrow y = x$. This makes clear that no cyclic permutation of a relation in S shares a subword of length 2 with a relation in H . Moreover, if we observe that two adjacent edges of a quasi-commutation 2-cell always determine the other two (as indicated in Fig. 29), we have that no cyclic permutation of a relation in S shares a subword of length 2 with another relation in S . We conclude that Condition **P2** is satisfied.
- Condition **P5** (fusion of heads) is trivially satisfied because the spiderheads are cyclic.

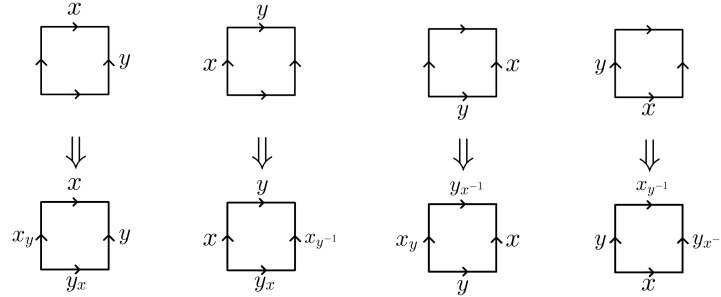


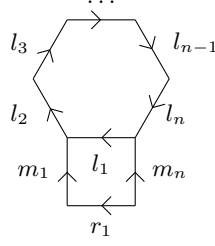
FIG. 29. Any two consecutive edges determine the rest.

In the process of writing this work, one of the authors in [4] brought to our attention that the cube completion in the Cayley complex of a trickle group can be deduced from the results in [4, Section 7] with regard to the sharp θ -cube condition. For the readers not familiar with the ideas from Garside theory that their results require, we include a proof solely based on the definition of trickle groups.

The rest of this section is dedicated to showing that conditions **P3** and **P4** are satisfied by a trickle group presentation, we start by looking at condition **P4** (prism completion), for which we need these short lemmas.

Lemma 6.4. *Let Γ be a trickle graph and let $x, y \in V(\Gamma)$. If $y < x$, then $y_x < x$.*

Proof. Notice that, if $y < x$, condition (a) ensures that $\{x, y\} \in E(\Gamma)$. By condition (d), we have that $x_x = x$. Now, if $y < x$, by condition (c) and the fact that φ_x is a bijection on $\text{Star}_\Gamma(x)$, we conclude that $y_x < x_x = x$. $\square$


 FIG. 30. Relations $\tau = l_1 \dots l_n$, $\sigma = l_1 m_1^{-1} r_1^{-1} m_n$ as in **P4**.

Lemma 6.5. *Let Γ be a trickle graph, let $\{x, y\} \in E(\Gamma)$ and let $\varepsilon \in \{-1, 1\}$. Then $y_{x^\varepsilon} = y_{(x_y)^\varepsilon}$ and, as a consequence, $y_{(x_{y_i})^\varepsilon} = y_{x^\varepsilon}$ for any i in $\mathbb{N}$.*

Proof. As $x \in \text{Star}(y)$, if $x \not\leq y$, then by condition (d) $x_y = x$ and, therefore, $y_{x^\varepsilon} = y_{(x_y)^\varepsilon}$. If $x < y$, by Lemma 6.4, we also have $x_y < y$ and, applying condition (d), we obtain $y_{x^\varepsilon} = y = y_{(x_y)^\varepsilon}$. $\square$

Proposition 6.6. *Let Γ be a trickle graph and let $\langle \Sigma \mid H \cup S \rangle$ be the presentation for the corresponding trickle group, where the partition of relations $H \cup S$ is done as described above. This presentation satisfies condition **P4** (prism completion) from Definition 2.6.*

Proof. Consider relations $\tau = l_1 \dots l_n$ in H and $\sigma = l_1 m_1^{-1} r_1^{-1} m_n$ in S as described in condition **P4** and depicted in Fig. 30. Note that, for a trickle group, there is a generator x in Σ with $\mu(x) = n$ such that $l_i = x$ for all $i = 1, \dots, n$, or $l_i = x^{-1}$ for all $i = 1, \dots, n$. Depending on whether m_n is labelled by a generator y in Σ or y^{-1} in Σ^{-1} , each of these two cases gives rise to two possible diagrams: a total of four cases which are depicted in Fig. 31, where the directed edges are labelled by $x, y \in \Sigma$. We argue why **P4** is satisfied in the cases depicted by Fig. 31(A) and

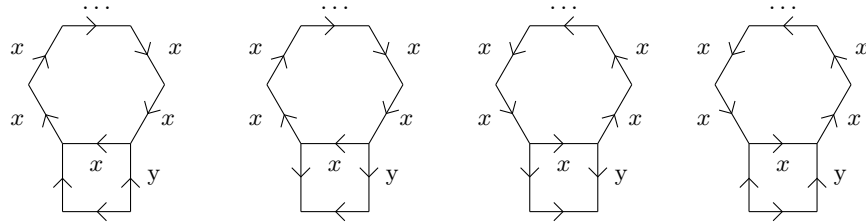
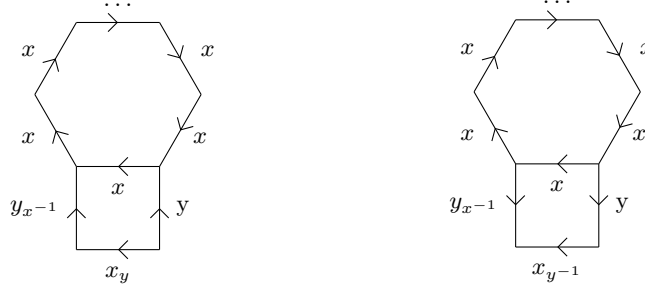

 (A) $l_1 = x, m_n = y$ (B) $l_1 = x, m_n = y^{-1}$ (C) $l_1 = x^{-1}, m_n = y$ (D) $l_1 = x^{-1}, m_n = y^{-1}$

 FIG. 31. The relations σ and τ for **P4** in a trickle group, depending on whether l_1 and m_n are labelled by a generator in Σ or its inverse.

Fig. 31(B), and cases Fig. 31(C) and Fig. 31(D) follow symmetrically. First note that, taking into account that two adjacent edges in a square determine the whole square (Fig. 29), the unlabelled edges in Fig. 31(A) and Fig. 31(B) can be written as shown in Fig. 32(A) and Fig. 32(B) respectively. Notice that $y_{x^{n-1}} = y_{x^{-1}}$ since $\mu(x) = n$ and the order of φ_x divides $\mu(x)$.



(A) $l_1 = x, m_n = y, r_1 = x_y, m_1 = y_{x^{-1}}$. (B) $l_1 = x, m_n = y^{-1}, r_1 = x_{y^{-1}}, m_1 = y_{x^{-1}}^{-1}$.

FIG. 32. The remaining labels for the edges of the square can be inferred from the labels in Fig. 31

We first focus on the case shown in Fig. 32(A), where σ_1 is given by the square

$$\sigma_1 = x(y_{x^{n-1}})^{-1}(x_y)^{-1}y$$

corresponding to a quasi-commutation between the generator x and the generator $y_{x^{n-1}}$. We claim that taking $r_i = x_y$ and $m_i = y_{x^{n-i}}$ for $i = 1, \dots, n$ yields the relations $\rho, \sigma_2, \dots, \sigma_n$ that fulfill condition **P4**. First, $\rho = r_1 \dots r_n = (x_y)^n$ clearly defines a relation in H since $\mu(x) = \mu(x_y)$ by the definition of a trickle graph. Hence, we only need to check that

$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1} = x(y_{x^{n-i}})^{-1}(x_y)^{-1}y_{x^{n-i+1}}$$

defines a relation in the trickle group, but this is the same as asking if

$$x_y y_{x^{n-i}} = y_{x^{n-i+1}} x.$$

Since σ_1 defines the relation $x_y y_{x^{n-1}} = y_x x$, we know that $\{x, y_{x^{n-1}}\} \in E(\Gamma)$ and therefore also $\{x, y_{x^{n-i}}\} \in E(\Gamma)$ for $i = 2, \dots, n$. The relation induced by this edge is, by definition,

$$x_{y_{x^{n-i}}} y_{x^{n-i}} = y_{x^{n-i}} x,$$

which becomes the desired relation after noting that $x_{y_{x^{n-i}}} = x_y$ (Lemma 6.5).

Having shown that prism completion is possible in the case of Fig. 32(A), we proceed similarly for the case of Fig. 32(B). In this case, σ_1 is given by the square

$$\sigma_1 = x y_{x^{n-1}} (x_{y^{-1}})^{-1} y^{-1}$$

coming from the quasi-commutation relation between the generators $x_{y^{-1}}$ and $y_{x^{n-1}}$. We will see that taking $r_i = x_{y^{-1}}$ and $m_i = (y_{x^{n-i}})^{-1}$ for $i = 1, \dots, n$ yields the relations $\rho, \sigma_2, \dots, \sigma_n$ that fulfill condition **P4**. As before, $\rho = r_1 \dots r_n = (x_{y^{-1}})^n$ defines a relation in H because $\mu(x) = \mu(x_{y^{-1}})$, so we only need to verify that

$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1} = x y_{x^{n-i}} (x_{y^{-1}})^{-1} (y_{x^{n-i+1}})^{-1}$$

defines a square relation in the trickle group, equivalently, we need to see that

$$x y_{x^{n-i}} = y_{x^{n-i+1}} x_{y^{-1}}.$$

But this is a quasi-commutation relation induced by $y_{x^{n-i}}$ and $x_{y^{-1}}$, which can be checked by taking into account that either $x_{y^k} = x$ or $y_{x^k} = y$ for all $k \in \mathbb{Z}$.

To ensure that this quasi-commutation is indeed a square relation in the trickle group, we need to check that $y_{x^{n-i}}$ and $x_{y^{-1}}$ are joined by an edge in Γ . This is a consequence of the fact that $x_{y^{-1}}$ and $y_{x^{-1}}$ share an edge in Γ , because this means that $(y_{x^{-1}})_{x^{n-i+1}}$ (which is equal to $y_{x^{n-i}}$) and $(x_{y^{-1}})_{x^{n-i+1}}$ share an edge in Γ , and $(x_{y^{-1}})_{x^{n-i+1}} = x_{y^{-1}}$ by Lemma 6.5. $\square$

Proving that a trickle group presentation satisfies condition **P3** (cube completion) requires a bit more work and we need to take into consideration the possible orientations of the edges in the squares of a half-cube.

Because the square 2-cells in a trickle group are associated to relations of the type $y_x x = x_y y$, opposite edges in these squares have the same orientation. This means that the orientation of the edges in the boundary of the half-cube defined by three squares σ_1 , σ_2 and σ_3 only depends on the orientation of the three edges that are incident to the unique vertex in the interior of the half-cube (each of these edges may be pointing towards or against this vertex). Thus, we may find four different types of half-cubes given by a trickle group presentation, which have been depicted in Figure 33.

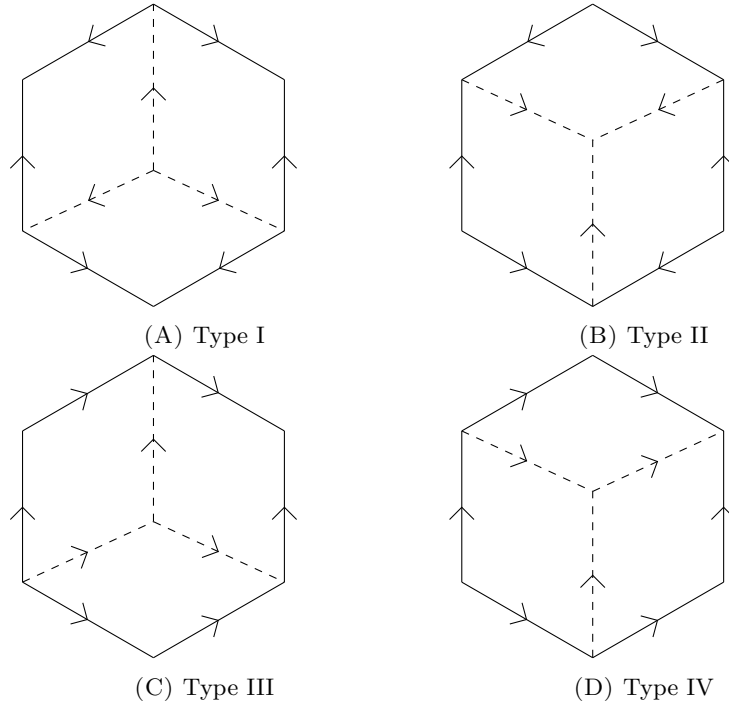


FIG. 33. Four types of half-cubes over trickle groups.

Notice that these half-cubes come in pairs: the underlying directed hexagon is identical for half-cubes of types I and II, and the same happens for those of types III and IV. We will show that a half-cube of type I in X_Γ admits a hexagonal move to a half-cube of type II, and vice versa; the same will be shown for types III and IV (as depicted in Figure 34). This, which is encompassed by Propositions 6.8, 6.11

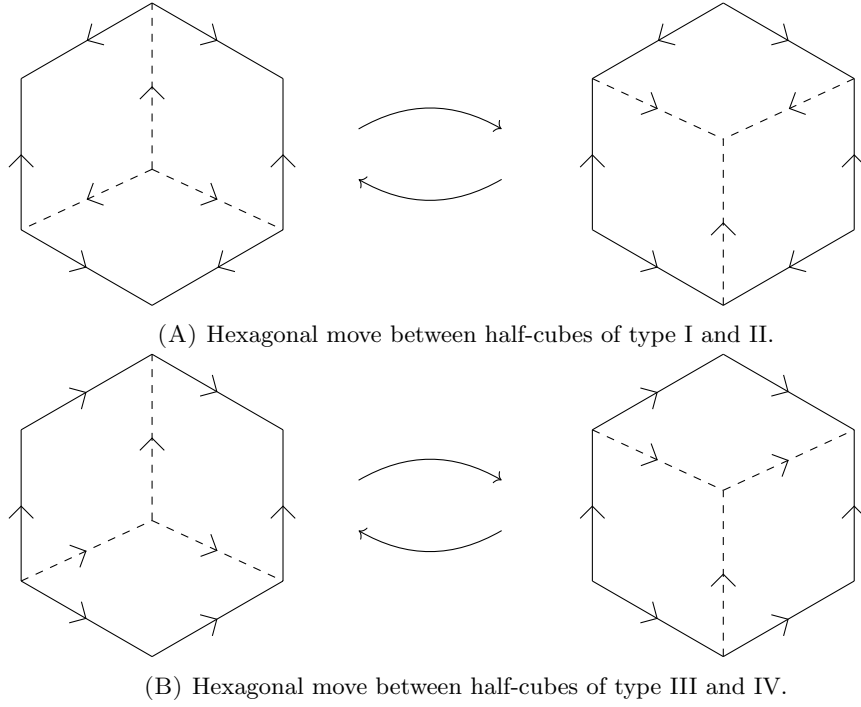


FIG. 34. The four types of half-cubes come in complementary pairs.

and 6.12, proves that condition **P3** (cube completion) in Definition 2.6 is satisfied by trickle groups.

Proposition 6.8 states that a hexagonal move can be applied to a half-cube of type II to obtain a half-cube of type I, and that a hexagonal move can be applied to a half-cube of type IV to a half-cube of type III. In order to prove this statement, Lemma 6.7 is needed to make some observations on how different generators of the group relate to each other.

Lemma 6.7. *Let Γ be a trickle graph and let $x, y, z \in V(\Gamma)$. If the vertices x, y and z are pairwise joined by edges in Γ , then the following equality holds*

$$(1) \quad z_{xyx} = z_{yx_y}.$$

Proof. Considering the possible order relations between the vertices, we need to verify the equality in several cases.

Case 1: Assume $x < z$. By condition (c), we also have $x_y < z_y$. Then, by condition (d), $z_x = z$ and $z_{yx_y} = (z_y)_{x_y} = z_y$, so the equality that we need to verify is $z_{yx} = z_y$.

(1.1) If $y < x$, using Lemma 6.4 and the transitivity of $\leq$, we have that $y < z$ and $y_x < z$, so $z_{yx} = z = z_y$ by condition (d).

(1.2) If $y \not< x$, condition (d) ensures that $y_x = y$ and we obtain $z_{yx} = z_y$.

Case 2: Assume $y < z$. This case is symmetrical to the previous one, so $z_{xyx} = z_{yx_y}$ holds.

Case 3: Assume $x \not< z$ and $y \not< z$.

- (3.1) If $x||y$, condition (d) yields $x_y = x$ and $y_x = y$, so the equality that we need to verify becomes $z_{xy} = z_{yx}$. Now, by condition (b), z cannot be smaller than both x and y , so it must be that either $x||z$ or $y||z$. Without loss of generality, suppose $y||z$. By condition (c), we have $y = y_x||z_x$ and, using condition (d), we obtain $z_{xy} = z_x = z_{yx}$.
- (3.2) If $y < x$, there are two possibilities:
- (3.2.1) If $z < y$, we have the chain $z < y < x$, so the equality $z_{yx} = z_{xy_x}$ is true by condition (g). Condition (d) gives $x_y = x$, so we have $z_{xy_x} = z_{yx} = z_{yx_y}$.
- (3.2.2) If $z \not< y$, since $y \not< z$, we must have $z||y$. And, since $x \not< z$, there are two subcases:
- If $z < x$, we will see that $z_{xy_x} = z_x = z_{yx_y}$. Indeed, from $z||y$, by condition (c), we have that $z_x||y_x$, which implies that $z_{xy_x} = z_x$ by condition (d). On the other hand, condition (d) ensures $z_y = z$ and $x_y = x$, so we have $z_{yx_y} = z_x$.
 - If $z||x$, we have that $z_{xy_x} = z_x = z_{yx_y}$. Indeed, condition (c) gives us that $z_x||y_x$, so $z_{xy_x} = z_x$ by condition (d). Now, because $z||y$ and $y < x$, condition (d) yields $z_{yx_y} = z_{x_y} = z_x$.
- (3.3) If $x < y$, the situation is analogous to the Case 3.2 due to the symmetry of the equation (1) with respect to x and y .

□

The above lemmas are all we need to prove that the “right to left” retessellations in Fig. 34, are possible:

Proposition 6.8 (II to I and IV to III). *Let Γ be a trickle graph.*

- (a) *If X_Γ contains a half-cube of type II, then it admits a hexagonal move that yields a half-cube of type I. And,*
- (b) *if X_Γ contains a half-cube of type IV, then it admits a hexagonal move that yields a half-cube of type III.*

Proof. (a) Consider the half-cube of type II and label the three incoming edges at the central vertex by $x, y, z \in V\Gamma$ as shown in Fig. 35(B). Since the three 2-cells come from quasi-commuting relations, their boundaries must be labelled by the relations

$$\begin{aligned} x_y y &= y_x x, \\ x_z z &= z_x x, \\ y_z z &= z_y y. \end{aligned}$$

By definition of the presentation, we have $\{x, y\}, \{x, z\}, \{y, z\} \in E\Gamma$. Firstly, this means that $x, y, z \in \text{Star}(x) \cap \text{Star}(y) \cup \text{Star}(z)$ and, since $\varphi_x, \varphi_y, \varphi_z$ are graph automorphisms, we also have that $\{x_z, y_z\}, \{x_y, z_y\}, \{y_x, z_x\} \in E\Gamma$ and so the following relations hold:

$$\begin{aligned} x_z y_z y_z &= y_z x_z x_z, \\ x_y z_y z_y &= z_y x_y x_y, \\ y_x z_x z_x &= z_x y_x y_x. \end{aligned}$$

Secondly, we are under the hypotheses of Lemma 6.7, so we have the equalities $x_z y_z = x_y z_y$, $y_z x_z = y_x z_x$ and $z_y x_y = z_x y_x$. As a consequence, the

2-cells given by the relations above can be glued in order to retessellate the half-cube (see Fig. 35(A)).

- (b) Consider the half-cube of type IV. Let v be the right top vertex. As shown in Fig. 35(D), label the three incoming edges at v by $x, y, z \in V\Gamma$, being z the only edge not in the boundary. The two squares that have v as one of its vertices come from quasi-commuting relations, so we have that $x, y \in \text{Star}(z)$ and that the boundaries of the squares must be labelled by the relations

$$\begin{aligned} z_x x &= x_z z, \\ z_y y &= y_z z. \end{aligned}$$

The remaining square also comes from a quasi-commuting relation, so its associated relation must be of the form

$$x_{zy} y_z = y_{zx} x_z.$$

From this relation, we see that $\{x_z, y_z\} \in E\Gamma$ and both vertices belong to $\text{Star}(z)$. As φ_z is a graph automorphism, we obtain that $\{x, y\} \in E\Gamma$ and therefore x, y, z are pairwise connected and the hypotheses of Lemma 6.7 hold. Because φ_x and φ_y are graph automorphisms, we get that $\varphi_x(\{y, z\}) = \{y_x, z_x\} \in E\Gamma$ and $\varphi_y(\{x, z\}) = \{x_y, z_y\} \in E\Gamma$. As we also have $\{x, y\} \in E\Gamma$, we obtain the following relations from the presentation:

$$\begin{aligned} x_y y &= y_x x, \\ y_x z_x z_x &= z_x y_x y_x, \\ x_y z_y z_y &= z_y x_y x_y. \end{aligned}$$

Now we can use these new relations to retessellate the half-cube using the equalities derived from Lemma 6.7, $x_{zy} = x_{yz_y}$, $y_{zx} = y_{xz_x}$ and $z_{xy} = z_{yx}$ (see Fig. 35(C)).

□

Propositions 6.11 and 6.12 prove that the remaining hexagonal moves (the “left-to-right” ones in Fig. 34) are also possible in trickle groups by showing that these are just inverse retessellations of the previous ones. Similar to before, in order to prove these statements Lemmas 6.9 and 6.10 make some preliminary observations relating the generators in the trickle group.

Lemma 6.9. *Let Γ be a trickle graph and let $\{x, y\} \in E(\Gamma)$. Then, $x_y < y_x$ if and only if $x < y$.*

Proof. ($\Rightarrow$) Suppose that $x \not< y$. Then, by condition (d), $x_y = x$ and we have that $x_x = x = x_y < y_x$ and, by condition (c), it must be that $x < y$, a contradiction. ($\Leftarrow$) If $x < y$, by condition (d), $y_x = y$. By Lemma 6.4, $x_y < y = y_x$. □

Lemma 6.10. *Let Γ be a trickle graph. If $\{x, y\} \in E\Gamma$, then $\{x, y_x\}, \{x_y, y\}, \{x_y, y_x\} \in E\Gamma$. In particular, the generators that label adjacent edges in a 2-cell of X_Γ correspond to adjacent vertices in Γ .*

Proof. Since φ_x and φ_y are graph automorphisms and $\{x, y\} \in E\Gamma$, we have that $\{x, y_x\} = \varphi_x(\{x, y\}) \in E\Gamma$ and $\{x_y, y\} = \varphi_y(\{x, y\}) \in E\Gamma$. To conclude, just recall that we always have either $x_y = x$ or $y_x = y$, so $\{x_y, y_x\} = \{x, y_x\} \in E\Gamma$ or $\{x_y, y_x\} = \{x_y, y\} \in E\Gamma$. □

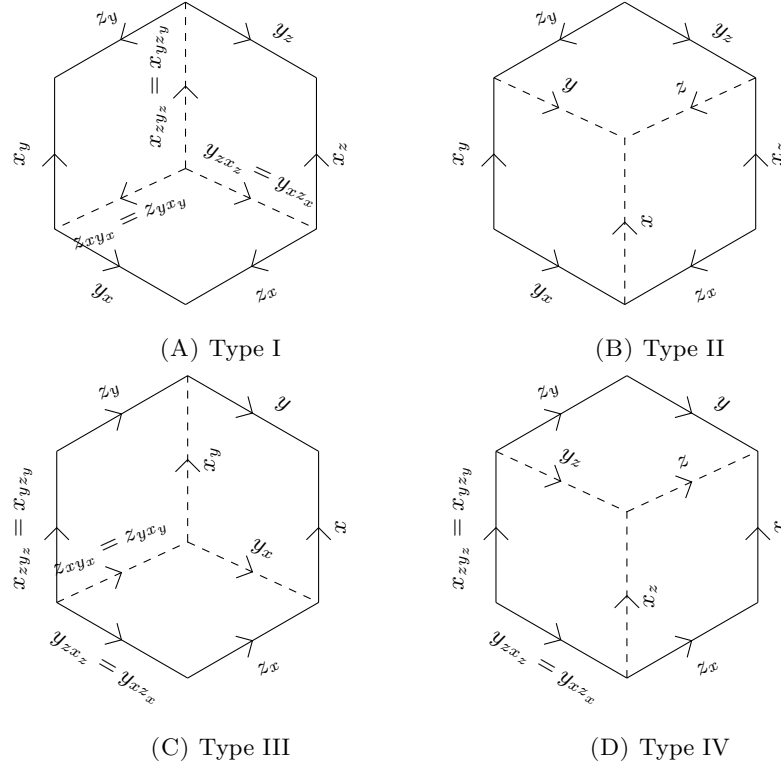


FIG. 35. Tessellated hexagons can always be labelled like this for some generators x, y, z .

Proposition 6.11 (Type III to type IV). *Let Γ be a trickle graph. If the standard presentation of the corresponding trickle group yields a half-cube of type III, then there exist pairwise adjacent vertices $x, y, z \in V(\Gamma)$ such that the edges of the tessellated hexagon admit the labelling in Figure 35(C).*

Proof. A half-cube of type III can always be labelled as shown in Figure 36, where $x, y, p, q \in V(\Gamma)$ satisfy that $\{x, y\}, \{p, x_y\}, \{q, y_x\} \in E(\Gamma)$ and $p_{x_y} = q_{y_x}$. Now, to show that we can have the labelling in Figure 35(C), it is enough to find $z \in V(\Gamma)$ adjacent to both x and y satisfying that $z_y = p$ and $z_x = q$.

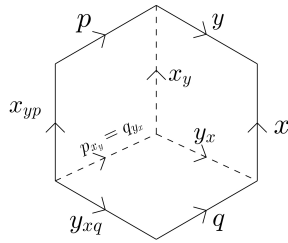


FIG. 36. Labelling for a type III hexagon.

We know that $\{q, y_x\} \in E\Gamma$ so, by Lemma 6.10, $\{q_{y_x}, y_x\} \in E\Gamma$. Since by hypothesis $p_{x_y} = q_{y_x}$ we have that $\{p_{x_y}, y_x\} \in E\Gamma$ and now, by Lemma 6.5, we see that $\{p_{x_y}, y_{x_y}\} \in E\Gamma$. Finally, since the automorphism $\varphi_{x_y}^{-1}$ preserves adjacency, we get that $\{p, y\} \in E\Gamma$. Since φ_{x_y} preserves adjacency, we have that $\{p, y\} \in E(\Gamma)$. A symmetric argument gives that $\{q, x\} \in E(\Gamma)$ as well. Now we can take $z', z'' \in V(\Gamma)$ such that $z'_y = p$ and $z''_x = q$. Clearly, z' is adjacent to y . Since $\{z'_y = p, x_y\} \in E(\Gamma)$ and φ_y preserves adjacency, z' is also adjacent to x . Thus, we may apply Lemma 6.7 to obtain $z'_{xy_x} = z'_{yx_y} = p_{x_y} = q_{y_x} = z''_{xy_x}$ and, because $\varphi_{y_x} \circ \varphi_x$ is injective, we conclude that $z' = z''$. Taking $z = z' = z''$ gives the desired labelling. $\square$

Proposition 6.12 (Type I to type II). *Let Γ be a trickle graph. If the standard presentation of the corresponding trickle group yields a half-cube of type I, then there exist pairwise adjacent vertices $x, y, z \in V(\Gamma)$ such that the edges of the tessellated hexagon admit the labelling in Figure 35(A).*

Proof. Given a half-cube of type I, there must be $p, q, r, s, u, v \in V(\Gamma)$ satisfying $u_p = v_r$, $p_u = q_s$ and $r_v = s_q$ so that edges can be labelled as in Figure 37. The goal now is to find pairwise adjacent vertices $x, y, z \in V(\Gamma)$ such that $u = x_y$, $v = x_z$, $r = y_z$, $s = y_x$, $p = z_y$ and $q = z_x$. We will do that at the very last stage of the proof.

We will begin showing that, without loss of generality, we can make some assumptions on the partial order restricted to $\{u_p, p_u, r_v\}$.

Notice that there are only five partial orders on a set of three elements $\{a, b, c\}$ up to order-isomorphism:

- the linear order: $a < b < c$,
- the order with a global minimum: $a < b, a < c$ and $b \parallel c$.
- the order with a global maximum: $a \parallel b, a < c$ and $b < c$.
- the order with a single chain of length 2: $b < c$, $a \parallel b$ and $a \parallel c$.
- the order where all elements are incomparable: $a \parallel b, b \parallel c$ and $a \parallel c$.

On the five cases, it always hold that a is smaller or incomparable to b and c and that b is smaller or incomparable to c . By the symmetry of the configuration in Fig. 37, we can assume, without loss of generality, that the following three facts hold:

- (1) u_p is smaller than or incomparable to r_v ,
- (2) u_p is smaller than or incomparable to p_u ,
- (3) r_v is smaller than or incomparable to p_u .

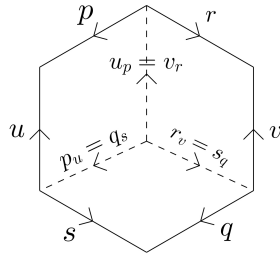


FIG. 37. Labelling for a half-cube of type I.

Notice that $p_u = q_s$ is greater than or incomparable to the other elements. In particular, we have that $u_p < p_u$ or $u_p || p_u$ which, by Lemma 6.9, implies that $u < p$ or $u || p$ and, by condition (d), we have that $p_u = p$. Similarly, starting from $r_v = s_q < q_s = p_u$ or $s_q || q_s$, we conclude that $q_s = q$. Therefore, we have that $p = p_u = q_s = q$. In particular,

$$(2) \quad p_s = p.$$

Observe as well that $u_p = v_r < r_v$ or $v_r || r_v$. Again, the previous argument gives us that $r = r_v = s_q$ and, as we have shown that $p = q$, we have that

$$(3) \quad s_p = r.$$

We claim that $\{u, s\} \in E\Gamma$ and $\{v, p\} \in E\Gamma$. Indeed, since they are labels of consecutive edges of a square 2-cell, u_p and $s_q = s_p$ are adjacent in Γ and since φ_p is a graph automorphism, we obtain that $\{u, s\} \in E(\Gamma)$. Also, since $r = r_v < p_u$ or $r = r_v || p_u$ and $p_u = p$, we conclude that either $r < p$ or $r || p$ and therefore $p_r = p$. Given that $\{v_r, p_r = p\} \in E(\Gamma)$ and that adjacency is preserved by φ_r , we see that $\{v, p = z\} \in E(\Gamma)$.

By the previous paragraph, we can take $x' = \varphi_s^{-1}(u)$ and $x'' = \varphi_p^{-1}(v)$ and, using (2) and (3) respectively, we have

$$u_p = x'_{sp} = x'_{sp_s} \text{ and } v_r = x''_{pr} = x''_{ps_p}.$$

Now, since $u_p = v_r$, we have that $x'_{sp_s} = x''_{pr_p}$. Observe that $\{x'_s = u, p = p_s\} \in E\Gamma$ and, as φ_s preserves adjacency, we have that $\{x', p\} \in E\Gamma$. As $x'_s \in \text{Star}(s)$, $\{x', s\} \in E\Gamma$. Finally, as $p = q$, we see that $\{p, s\} \in E\Gamma$. Since we have checked that p, s, x' are pairwise adjacent, we may use Lemma 6.7 and get that $x'_{sp_s} = x'_{ps_p}$. Therefore, as $u_p = v_r$, we have $x'_{ps_p} = x''_{ps_p}$ and now the injectivity of the vertex automorphisms gives us that $x' = x''$.

Set $x = x' = x''$, $y = s$ and $z = p$. Notice that by the previous paragraph, we know that x, y, z are pairwise adjacent. We now proceed to verify that $u = x_y$, $v = x_z$, $r = y_z$, $s = y_x$, $p = z_y$ and $q = z_x$. First, we saw that $u = x'_s = x_y$. Second, we saw that $v = x''_p = x_z$. Third, by Eq. (3), we have that $r = y_z$. Fourth, because $u_q < s_q$ or $u_q || s_q$, using condition (c), we have that $u < s$ or $u || s$ and, using that $u = x_s$, $s = s_s$ and applying condition (c), we deduce that $x < s$ or $x || s$. So, by condition (d), $y_x = s_x = s$. Fifth, by Eq. (2), we have that $z_y = p$. Sixth, because $u_p < p_u$ or $u_p || p_u$, we have that $x_{sp} < p_{sp}$ or $x_{sp} || p_{sp}$ and condition (c) tells us that $x < p$ or $x || p$. And, by condition (d), $z_x = p_x = p = q$. $\square$

As a consequence of Propositions 6.8, 6.11 and 6.12, we have:

Proposition 6.13. *The standard presentation of a trickle group $\text{Tr}(\Gamma)$ satisfies condition P3 in Definition 2.6.*

Propositions 6.6 and 6.13, together with the observations made at the beginning of this section, allow us to conclude:

Theorem 6.14. *The standard presentation of a trickle group $\text{Tr}(\Gamma)$ is a prismatic presentation.*

6.2. Twisted right-angled Artin groups. This subsection is devoted to showing that the standard presentation for a T-RAAG is a prismatic presentation, which will provide examples different from trickle groups.

Remark 6.15. Twisted right-angled Artin groups are not trickle groups. In [13], the author shows that trickle groups where all the generators have infinite order must be orderable, but it is known by the results provided in [2] that there exist T-RAAGs which are not orderable.

Let Γ be a mixed graph and let $\langle \sigma | R \rangle$ be the standard presentation of a T-RAAG that is determined by Γ . To see that this is a prismatic presentation, we show that **P1-P5** hold.

To show that **P1** holds, it is enough to take $H = \emptyset$ and $S = R$, since all relators are of the form $aba^{-1}b^{-1}$ or $abab^{-1}$, which are clearly cyclically reduced words of length four.

To check **P2**, observe that any two adjacent edges of a 2-cell coming from a relator in R are labelled by different generators, so they determine the pair of edges in Γ that gives rise to the relator associated to the 2-cell. Since each pair of vertices in Γ may be joined by a unique edge (directed or undirected), it is impossible to find two different squares with two adjacent edges having the same labelling. Therefore, no cyclic permutation of a square relation shares a subword of length two with another relation.

Let us now verify **P3**. Suppose $\sigma_1, \sigma_2, \sigma_3$ are square relations that factorise as in **P3**. By looking at the generators that appear in the relation, we know which vertices a and b of Γ give rise to this relation and, depending on the shape of the relation ($aba^{-1}b^{-1}$ or $abab^{-1}$), we know if a and b are joined by a directed or undirected edge and, in the directed case, whether the edge points at a or at b . This determines a triangle in Γ that can only be of one of the seven types indicated in Fig. 38.

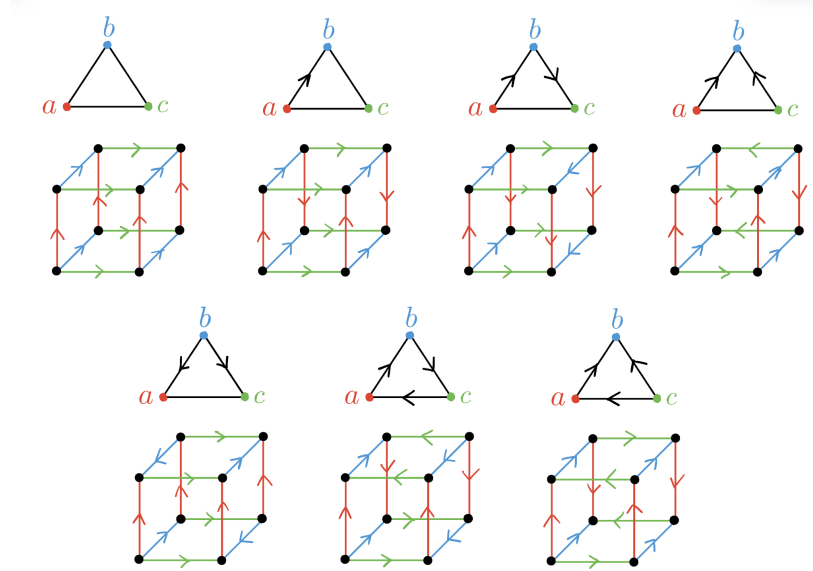


FIG. 38. Cubes in the Cayley complex of a T-RAAG.

For each of these triangles, we have cubes in the Cayley complex of the T-RAAG associated to Γ as in Fig. 38. In order to obtain the square relations τ_1, τ_2, τ_3 in

P3, we simply take the relations associated to the squares in the complement of $\sigma_1, \sigma_2, \sigma_3$ in the corresponding cube. Thus, **P3** is satisfied.

As for conditions **P4** and **P5**, they vacuously hold because we have taken H to be the empty subset. This discussion leads us to the following result:

Theorem 6.16. *Let Γ be a mixed graph. The T-RAAG associated to Γ admits a prismatic presentation.*

It is possible to find examples of prismatic presentations among torsion T-RAAGs, but restrictions must be put on the labelling of the vertices of the corresponding decorated mixed graph.

Corollary 6.17. *Let $\Gamma = (V, E, D, \iota, \tau, \mu)$ be a decorated mixed graph where the labelling of the vertices satisfies that, for every $[u, v]$ $v \in D$, $\mu(v)$ is even or $\mu(v) = \infty$.*

Proof. To show that **P1** holds, it is enough to take $H = \{v^{\mu(v)} : v \in V\Gamma \text{ with } \mu(v) \neq \infty\}$ and S as its complement in the set of relators of the presentation. Notice that S is the same set of relators that makes the presentation of the underlying T-RAAG prismatic, so it is clear by the arguments above that **P3** is satisfied. Let us check the other conditions.

For **P2**, observe that consecutive edges of a head have the same label, which does not happen for any pair of consecutive edges in a square, so no head can be glued along two consecutive edges to a square. The same is true for two squares, by the same argument we used for T-RAAGs.

To show that **P4** holds, let τ be a head and let σ_1 be a square sharing an edge labelled by $v \in V\Gamma$ with τ . Let $u \neq v$ the vertex that labels the other edges of σ_1 . By definition $\{u, v\} \in E\Gamma$ and we have several possibilities.

- The edge is of type $[u, v]$. Then one can build a prism gluing $\mu(v)$ copies of σ_1 and two copies of τ . Observe that the two bases of the prism will have parallel orientations.
- The edge is of type $[u, v]$. Then one can build a prism gluing $\mu(v)$ copies of σ_1 and two copies of τ . Notice that this is only possible because $\mu(v)$ is even (the edges joining the two bases of the prism present alternate orientations). In this case the two bases have parallel orientations as well.
- The edge is of type $[v, u]$. Then one can build a prism gluing $\mu(v)$ copies of σ_1 and two copies of τ . Observe that the two bases of the prism will have opposite orientations.

Finally, to verify **P5**, just notice that two heads that share an edge must be induced by the same relation so the cyclic reduction of the concatenation of the corresponding relators is the empty word.

□

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